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Medium conveyance device, control method, and control program

Abstract

A medium conveying apparatus includes a loading tray to place media, a feed roller to separate and sequentially feed the media placed on the loading tray, a motor to drive the feed roller, a sensor located on a downstream side of the feed roller in a medium conveying direction to detect a medium, and a processor to control the motor. In a separation period from a start of medium feeding by the feed roller to detection of a front edge of the first medium by the sensor, when feeding a first medium of a plurality of media placed on the loading tray, the processor controls the motor to rotate the feed roller at a constant speed, and when feeding a second or subsequent medium, the processor controls the motor to rotate the feed roller at a first speed and then at a second speed higher than the first speed.

Inventors: Shimosaka; Kiichiro (Kahoku, JP), Morikawa; Syuichi (Kahoku, JP)

Applicant: PFU LIMITED (Kahoku, JP)

Family ID: 84323977

Assignee: PFU LIMITED (Kahoku, JP)

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Primary Examiner: Gokhale; Prasad V

Attorney, Agent or Firm: WOMBLE BOND DICKINSON (US) LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of International Application No. PCT/JP2021/020730, filed on May 31, 2021, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates to a medium conveying apparatus, a control method, and a control program and particularly relates to a medium conveying apparatus, a control method, and a control program for separating and sequentially feeding media.

BACKGROUND

(2) In a medium conveying apparatus such as a scanner that separates and sequentially feeds a plurality of media to image, it is required to further reduce the time required for medium feeding. On the other hand, it is required for the medium conveying apparatus to temporarily stop medium feeding, for example, when there is insufficient free space in a storage device to store images of the media or when the distance between successively fed media is short. In such a case, the medium conveying apparatus is required to suitably control stoppage and resumption of medium feeding.

(3) A sheet feeding device to control a feed roller to a low speed, when feeding of a succeeding sheet by the feed roller is started after arrival of the rear edge of a preceding sheet is detected by a post-registration sensor, is disclosed (see International Application Publication No. WO 2019/130648). The sheet feeding device controls the feed roller to a high speed when the front edge of the succeeding sheet passes a nip position of the feed roller and a separation roller.

SUMMARY

(4) According to some embodiments, a medium conveying apparatus includes a loading tray to place media, a feed roller to separate and sequentially feed the media placed on the loading tray, a motor to drive the feed roller, a sensor located on a downstream side of the feed roller in a medium conveying direction to detect a medium, and a processor to control the motor. In a separation period from a start of medium feeding by the feed roller to detection of a front edge of the first medium by the sensor, when feeding a first medium of a plurality of media placed on the loading tray, the control module controls the motor to rotate the feed roller at a constant speed, and when feeding a second or subsequent medium, the control module controls the motor to rotate the feed roller at a first speed and then at a second speed higher than the first speed.

(5) According to some embodiments, a method for conveying a medium includes separating and sequentially feeding media placed on a loading tray by a feed roller, driving the feed roller by a motor, detecting a medium by a sensor located on a downstream side of the feed roller in a medium

conveying direction, and controlling the motor. In a separation period from a start of medium feeding by the feed roller to detection of a front edge of the first medium by the sensor, when feeding a first medium of a plurality of media placed on the loading tray, controlling the motor to rotate the feed roller at a constant speed, and when feeding a second or subsequent medium, controlling the motor to rotate the feed roller at a first speed and then at a second speed higher than the first speed.

(6) According to some embodiments, a computer-readable, non-transitory medium storing executable instructions for conveying a medium is provided. The executable instructions includes controlling a motor to drive a feed roller to separate and sequentially feed media placed on a loading tray. In a separation period from a start of medium feeding by the feed roller to detection of a front edge of the first medium by a sensor located on a downstream side of the feed roller in a medium conveying direction, when feeding a first medium of media placed on the loading tray, controlling the motor to rotate the feed roller at a constant speed, and when feeding a second or subsequent medium, controlling the motor to rotate the feed roller at a first speed and then at a second speed higher than the first speed.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a perspective view illustrating an example of a medium conveying apparatus according to an embodiment.

(2) FIG. 2 is a diagram for illustrating an example of a conveyance path inside a medium conveying apparatus.

(3) FIG. 3 is a schematic diagram for illustrating examples of driving sources.

(4) FIG. 4 is a schematic diagram for illustrating an example of third medium sensor, etc.

(5) FIG. 5 is a schematic diagram for illustrating an example of a third medium sensor, etc.

(6) FIG. 6 is a block diagram illustrating a schematic configuration of an example of a medium conveying apparatus.

(7) FIG. 7 is a diagram illustrating a schematic configuration of an example of a storage device and a processing circuit.

(8) FIG. 8 is a flowchart illustrating an operation example of medium reading processing.

(9) FIG. 9 is a flowchart illustrating an operation example of the medium reading processing.

(10) FIG. 10 is a graph for illustrating changes in the speeds of rollers.

(11) FIG. 11 is a graph for illustrating changes in the speeds of the rollers.

(12) FIG. 12 is a graph for illustrating changes in the speeds of the rollers.

(13) FIG. 13A is a schematic diagram for illustrating technical significance.

(14) FIG. 13B is a schematic diagram for illustrating technical significance.

(15) FIG. 14 is a schematic diagram for illustrating technical significance.

(16) FIG. 15 is a schematic diagram for illustrating other examples of driving sources.

(17) FIG. 16 is a flowchart illustrating an example of part of operation of another type of medium reading processing.

(18) FIG. 17 is a graph for illustrating changes in the speed of a roller.

(19) FIG. 18 is a flowchart illustrating an example of part of operation of yet another type of medium reading processing.

(20) FIG. 19 is a graph for illustrating changes in the speed of a roller.

(21) FIG. 20 is a flowchart illustrating an operation example of yet another type of medium reading processing.

(22) FIG. 21 is a flowchart illustrating an operation example of the yet another type of medium reading processing.

- (23) FIG. 22 is a graph for illustrating changes in the speed of a roller.
- (24) FIG. 23 is a schematic diagram for illustrating yet other driving sources.
- (25) FIG. 24 is a flowchart illustrating an operation example of yet another type of medium reading processing.
- (26) FIG. 25 is a flowchart illustrating an operation example of yet another type of medium reading processing.
- (27) FIG. 26 is a graph for illustrating changes in the speeds of rollers.
- (28) FIG. 27 is a schematic diagram for illustrating yet other driving sources.
- (29) FIG. 28 is a schematic diagram for illustrating yet other driving sources.
- (30) FIG. 29 is a flowchart illustrating an example of part of operation of yet another type of medium reading processing.
- (31) FIG. 30 is a graph for illustrating changes in the speeds of rollers.
- (32) FIG. 31 is a schematic diagram for illustrating yet other driving sources.
- (33) FIG. 32 is a flowchart illustrating an example of part of operation of yet another type of medium reading processing.
- (34) FIG. 33 is a graph for illustrating changes in the speeds of rollers.
- (35) FIG. 34 is a flowchart illustrating an operation example of yet another type of medium reading processing.
- (36) FIG. 35 is a flowchart illustrating an operation example of yet another type of medium reading processing.
- (37) FIG. 36 is a graph for illustrating changes in the speeds of rollers.
- (38) FIG. 37 is a diagram illustrating a schematic configuration of another example of processing circuit.

DESCRIPTION OF EMBODIMENTS

- (39) The object and advantages of the invention will be realized and attained by means of the elements and combinations, in particular, described in the claims. It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory, and are not restrictive of the invention as claimed.
- (40) Hereinafter, a medium conveying apparatus, a control method and a control program according to an embodiment, will be described with reference to the drawings. However, it should be noted that the technical scope of the invention is not limited to these embodiments, and extends to the inventions described in the claims and their equivalents.
- (41) FIG. 1 is a perspective view illustrating an example of a medium conveying apparatus configured as an image scanner. The medium conveying apparatus **100** conveys and images a medium being a document. Examples of a medium include paper, thick paper, a card, a booklet, and a passport. The medium conveying apparatus **100** may be a facsimile, a copying machine, a multifunctional peripheral (MFP), etc. A conveyed medium may be an object being printed on, etc., instead of a document, and the medium conveying apparatus **100** may be a printer, etc.
- (42) The medium conveying apparatus **100** includes a lower housing **101**, an upper housing **102**, a loading tray **103**, an ejection tray **104**, an operation device **105**, a display device **106**, etc.
- (43) The upper housing **102** is located at a position covering the top surface of the medium conveying apparatus **100** and is engaged with the lower housing **101** by a hinge to be openable when, for example, a medium is stuck or cleaning of the inside of the medium conveying apparatus **100** is performed.
- (44) The loading tray **103** is engaged with the lower housing **101** and places a medium to be fed and conveyed. The ejection tray **104** is engaged with the upper housing **102** and places an ejected medium. The ejection tray **104** may be engaged with the lower housing **101**.
- (45) The operation device **105** includes an input device such as a button, and an interface circuit acquiring a signal from the input device, accepts an input operation by a user, and outputs an operation signal based on the input operation by the user. The display device **106** includes a display

including a liquid crystal, an organic electro-luminescence (EL), etc., and an interface circuit outputting image data to the display, and displays the image data on the display.

(46) FIG. 2 is a diagram for illustrating a conveyance path inside the medium conveying apparatus **100**.

(47) The conveyance path inside the medium conveying apparatus **100** includes a first medium sensor **111**, a feed roller **112**, a brake roller **113**, a second medium sensor **114**, an ultrasonic sensor **115**, a third medium sensor **116**, a fourth medium sensor **117**, a fifth medium sensor **118**, a conveyance roller **119**, a first facing roller **120**, a sixth medium sensor **121**, an imaging device **122**, an ejection roller **123**, a second facing roller **124**, etc.

(48) Each of the numbers of the feed roller **113**, the brake roller **114**, the conveyance roller **119**, the first facing roller **120**, the ejection roller **123**, and/or the second facing roller **124** is not limited to one and may be more than one. In that case, a plurality of feed rollers **113**, brake rollers **114**, conveyance rollers **119**, first facing rollers **120**, ejection rollers **123**, and/or second facing rollers **124** are respectively spaced in the width direction A2 perpendicular to the medium conveying direction A1.

(49) The top surface of the lower housing **101** forms a lower guide **101a** of the conveyance path of a medium, and the bottom surface of the upper housing **102** forms an upper guide **102a** of the conveyance path of a medium. An arrow A1 in FIG. 2 indicates a medium conveying direction. Hereinafter, an upper stream refers to an upper stream in the medium conveying direction A1, and a lower stream refers to a lower stream in the medium conveying direction A1.

(50) The first medium sensor **111** is located on the upstream side of the feed roller **112** and the brake roller **113**. The first medium sensor **111** includes a contact detection sensor and detects whether a medium is placed on the loading tray **103**. The first medium sensor **111** generates and outputs a first medium signal, the signal value of which varies between a state in which a medium is placed on the loading tray **103** and a state in which a medium is not placed. The first medium sensor **111** is not limited to a contact detection sensor and any other sensor that can detect existence of a medium, such as a light detection sensor, may be used as the first medium sensor **111**.

(51) The feed roller **112** is provided in the lower housing **101**, separates and sequentially feeds media placed on the loading tray **103** from the lower side. The brake roller **113** is provided in the upper housing **102**, to face the feed roller **112**, and rotates in a direction opposite to a medium feeding direction. The feed roller **112** may be provided in the upper housing **102**, the brake roller **113** may be provided in the lower housing **101**, and the feed roller **112** may sequentially feed media placed on the loading tray **103** from the upper side.

(52) The second medium sensor **114** is an example of a second sensor, is located on the downstream side of the feed roller **112** and on the upstream side of the conveyance roller **119**, and detects a medium conveyed to the position of the sensor. In particular, the second medium sensor **114** is located between the feed roller **112** and the fifth medium sensor **118** in the medium conveying direction A1 and close to a nip region of the feed roller **112** and the brake roller **113**. The second medium sensor **114** includes a light emitter and a light receiver that are provided on one side of the medium conveyance path and a light-guiding tube, provided at a position facing the light emitter and the light receiver with the medium conveyance path in between. The light emitter is a light emitting diode (LED), etc., and emits light toward the medium conveyance path. The light receiver is a photodiode, etc., and receives light emitted by the light emitter and guided by the light guide. When a medium is present at a position facing the second medium sensor **114**, light emitted from the light emitter is blocked by the medium, and therefore the light receiver does not detect the light emitted from the light emitter. The second medium sensor **114** generates and outputs a second medium signal, the signal value of which varies between a state in which a medium is present at the position of the second medium sensor **114** and a state in which a medium is not present, based on the intensity of light received by the light receiver.

(53) The ultrasonic sensor **115** is an example of a thickness detection sensor. The ultrasonic sensor

115 is located on the downstream side of the feed roller **112** and on the upstream side of the conveyance roller **119**. The ultrasonic sensor **115** includes an ultrasonic transmitter **115a** and an ultrasonic receiver **115b**. The ultrasonic transmitter **115a** and the ultrasonic receiver **115b** are located close to the conveyance path of a medium to face each other with the conveyance path in between. The ultrasonic transmitter **115a** transmits an ultrasonic wave. The ultrasonic receiver **115b** receives an ultrasonic wave transmitted by the ultrasonic transmitter **115a** and passing through a medium and generates and outputs an ultrasonic signal being an electric signal based on the received ultrasonic wave. When a plurality of media are conveyed in a stacked manner, an ultrasonic wave passing through the media is attenuated by an air layer between the media conveyed in a stacked manner. Accordingly, the medium ejection apparatus **100** can detect multi feed of media, based on the ultrasonic signal. Further, an ultrasonic wave passing through a medium is also attenuated by the medium itself, and an amount of attenuation increases as the thickness of the medium through which the ultrasonic wave passes increases. Accordingly, the medium ejection apparatus **100** can detect the thickness of the conveyed medium, based on the ultrasonic signal.

(54) The fifth medium sensor **118** is an example of a sensor, is located on the downstream side of the feed roller **112** and on the upstream side of the conveyance roller **119**, and detects a medium conveyed to the position of the sensor. In other words, the fifth medium sensor **118** is located between the feed roller **112** and the conveyance roller **119**. The fifth medium sensor **118** includes a light emitter and a light receiver that are provided on one side of the medium conveyance path and a light-guiding tube, provided at a position facing the light emitter and the light receiver with the medium conveyance path in between. The light emitter is an LED, etc., and emits light toward the medium conveyance path. The light receiver is a photodiode, etc., and receives light emitted by the light emitter and guided by the light-guiding tube. The fifth medium sensor **118** generates and outputs a fifth medium signal, the signal value of which varies between a state in which a medium is present at the position of the fifth medium sensor **118** and a state in which a medium is not present, based on the intensity of light received by the light receiver.

(55) The conveyance roller **119** and the first facing roller **120** are located on the downstream side of the feed roller **112** to face each other and convey a medium fed by the feed roller **112** and the brake roller **113** to the imaging device **122**. The conveyance roller **119** is provided in the upper housing **102**, and the first facing roller **120** is provided on the lower side of the conveyance roller **119** in the lower housing **101**.

(56) The sixth medium sensor **121** is located on the downstream side of the conveyance roller **119** and on the upstream side of the imaging device **122** and detects a medium conveyed to the position of the sensor. The sixth medium sensor **121** includes a light emitter and a light receiver that are provided on one side of the medium conveyance path and a lightguide, provided at a position facing the light emitter and the light receiver with the medium conveyance path in between. The light emitter is an LED, etc., and emits light toward the medium conveyance path. The light receiver is a photodiode, etc., and receives light emitted by the light emitter and guided by the light guide. The sixth medium sensor **121** generates and outputs a sixth medium signal, the signal value of which varies between a state in which a medium is present at the position of the sixth medium sensor **121** and a state in which a medium is not present, based on the intensity of light received by the light receiver.

(57) The imaging device **122** is an example of an imaging module, is located on the downstream side of the conveyance roller **119**, and images a medium conveyed by the conveyance roller **119**. The imaging device **122** includes a first imaging device **122a** and a second imaging device **122b** that are located to face each other with the medium conveyance path in between. The first imaging device **122a** includes a line sensor based on a unity-magnification optical system type contact image sensor (CIS) including complementary metal oxide semiconductor- (CMOS-)based imaging elements linearly arranged in a main scanning direction. The first imaging device **122a** further

includes lenses each forming an image on an imaging element, and an A/D converter amplifying and analog-digital (A/D) converting an electric signal output from the imaging element. The first imaging device **122a** generates an input image by imaging the front side of a conveyed medium in accordance with control from a processing circuit described later and outputs the generated image. (58) The second imaging device **122b** includes a line sensor based on a unity-magnification optical system type CIS including CMOS-based imaging elements linearly arranged in the main scanning direction. The second imaging device **122b** further includes lenses each forming an image on an imaging element, and an A/D converter amplifying and analog-digital (A/D) converting an electric signal output from the imaging element. The second imaging device **122b** generates an input image by imaging the back side of a conveyed medium in accordance with control from the processing circuit described later and outputs the generated image.

(59) Only one of the first imaging device **122a** and the second imaging device **122b** may be located and only one side of a medium may be read in the medium conveying apparatus **100**. Further, a line sensor based on a unity-magnification optical system type CIS including charge coupled device-(CCD-)based imaging elements may be used in place of the line sensor based on a unity-magnification optical system type CIS including CMOS-based imaging elements. Further, a reduction optical system type line sensor including CMOS-based or CCD-based imaging elements may be used. The image sensors may be located to face each other, or may not be located to face each other.

(60) The ejection roller **123** and the second facing roller **124** are located on the downstream side of the imaging device **122** to face each other and eject a medium conveyed by the conveyance roller **119** and the first facing roller **120** and imaged by the imaging device **122** into the ejection tray **104**. The ejection roller **123** is provided in the upper housing **102**, and the second facing roller **124** is provided on the lower side of the ejection roller **123** in the lower housing **101**.

(61) A medium placed on the loading tray **103** is conveyed between the lower guide **101a** and the upper guide **102a** toward the medium conveying direction A1 by the feed roller **112** rotating in a direction of an arrow A3 in FIG. 2, i.e., the medium ejecting direction. The brake roller **113** rotates in a direction of an arrow A3, i.e., a direction opposite to the medium feeding direction at the time of medium conveying. When a plurality of media are placed on the loading tray **103**, only a medium in contact with the feed roller **112** of the medium placed on the loading tray **103** is separated by working of the feed roller **112** and the brake roller **113**. Consequently, conveyance of a medium other than the separated medium is restricted (prevention of multi feed). The brake roller **113** may separate the medium by stopping. A separation pad may be used in place of the brake roller **113**.

(62) A medium is fed between the conveyance roller **119** and the first facing roller **120** while being guided by the lower guide **101a** and the upper guide **102a**. The medium is fed between the first imaging device **122a** and the second imaging device **122b** by the conveyance roller **119** and the first facing roller **120** rotating in directions of an arrow A4 and an arrow A5, respectively. The medium read by the imaging device **122** is ejected into the ejection tray **104** by the ejection roller **123** and the second facing roller **124** rotating in directions of an arrow A6 and an arrow A7, respectively.

(63) FIG. 3 is a schematic diagram for illustrating driving sources of the feed roller **112**, the brake roller **113**, the conveyance roller **119**, the first facing roller **120**, the ejection roller **123**, and/or the second facing roller **124**.

(64) As illustrated in FIG. 3, the medium conveying apparatus **100** includes a first motor **131** and a second motor **132** as driving sources of the rollers.

(65) The first motor **131** is an example of a motor, is provided in the lower housing **101**, is connected to the feed roller **112** through a first transmission mechanism **131a**, and drives the feed roller **112**. The first motor **131** generates a driving force for driving the feed roller **112** in accordance with a control signal from a processing circuit. The first transmission mechanism **131a**

includes one or a plurality of pulleys, belts, gears, etc., that are provided between the first motor **131** and a shaft **112a** of the feed roller **112** and transmits the driving force generated by the first motor **131** to the feed roller **112**. The first motor **131** feeds a medium by rotating the feed roller **112**.

(66) The second motor **132** is provided in the upper housing **102** separately from the first motor **131**, is connected to the conveyance roller **119**, the ejection roller **123**, and the brake roller **113** through a second transmission mechanism **132a**, and drives the conveyance roller **119**, the ejection roller **123**, and the brake roller **113**. The second motor **132** generates a driving force for driving the conveyance roller **119**, the ejection roller **123**, and the brake roller **113** in accordance with a control signal from the processing circuit. The second transmission mechanism **132a** includes one or a plurality of pulleys, belts, gears, etc., that are provided between the second motor **132**, and a shaft **119a** of the conveyance roller **119**, a shaft **123a** of the ejection roller **123**, and a shaft **113a** of the brake roller **113**. In particular, one or a plurality of gears for changing the rotation direction and the rotation speed of each roller are provided between the shaft **119a** of the conveyance roller **119** and/or the shaft **123a** of the ejection roller **123**, and the shaft **113a** of the brake roller **113**. The second transmission mechanism **132a** transmits the driving force generated by the second motor **132** to the conveyance roller **119**, the ejection roller **123**, and the brake roller **113**. The second motor **132** causes the conveyance roller **119**, the ejection roller **123**, and the brake roller **113** to feed, convey, and eject a medium by rotating the conveyance roller **119**, the ejection roller **123**, and the brake roller **113**. The second motor **132** is an example of a driving source of the brake roller **113**.

(67) The first facing roller **120** is a driven roller driven to rotate by the conveyance roller **119**, and the second facing roller **124** is a driven roller driven to rotate by the ejection roller **123**. The first facing roller **120** and/or the second facing roller **124** may be provided to be driven by the driving force from the second motor **132**. In that case, one or a plurality of gears are further provided between the shaft **119a** of the conveyance roller **119** and the shaft **120a** of the first facing roller **120** and/or between the shaft **123a** of the ejection roller **123** and the shaft **124a** of the second facing roller **124**. The second transmission mechanism **132a** further transmits the driving force generated by the second motor **132** to the first facing roller **120** and/or the second facing roller **124**.

(68) FIG. 4 and FIG. 5 are schematic diagrams for illustrating the third medium sensor **116** and the fourth medium sensor **117**. FIG. 4 is a schematic diagram of the lower guide **101a** of the lower housing **101** viewed from above, and FIG. 5 is a schematic diagram of the upper guide **102a** of the upper housing **102** viewed from below.

(69) As illustrated in FIG. 4 and FIG. 5, the third medium sensor **116** and the fourth medium sensor **117** are located on the downstream side of the feed roller **112** and on the upstream side of the conveyance roller **119** and detect a medium conveyed to the positions of the sensors. In particular, the third medium sensor **116** and the fourth medium sensor **117** are located between the second medium sensor **114** and the fifth medium sensor **118** in the medium conveying direction **A1**. The third medium sensor **116** and the fourth medium sensor **117** may be located at positions almost identical to that of the fifth medium sensor **118** in the medium conveying direction **A1**. Further, the third medium sensor **116** and the fourth medium sensor **117** are spaced in a width direction **A8** perpendicular to the medium conveying direction.

(70) The third medium sensor **116** includes a light emitter and a light receiver that are provided on one side of the medium conveyance path and a light guide, provided at a position facing the light emitter and the light receiver with the medium conveyance path in between. The light emitter is an LED, etc., and emits light toward the medium conveyance path. The light receiver is a photodiode, etc., and receives light emitted by the light emitter and guided by the light guide. The third medium sensor **116** generates and outputs a third medium signal, the signal value of which varies between a state in which a medium is present at the position of the third medium sensor **116** and a state in which a medium is not present, based on the intensity of light received by the light receiver.

(71) The fourth medium sensor **117** includes a light emitter and a light receiver that are provided on one side of the medium conveyance path and a light guide, provided at a position facing the light emitter and the light receiver with the medium conveyance path in between. The light emitter is an LED, etc., and emits light toward the medium conveyance path. the light receiver is a photodiode, etc., and receives light emitted by the light emitter and guided by the light guide. The fourth medium sensor **117** generates and outputs a fourth medium signal, the signal value of which varies between a state in which a medium is present at the position of the fourth medium sensor **117** and a state in which a medium is not present, based on the intensity of light received by the light receiver.

(72) A reflection member such as a mirror may be used in place of the light guide in the second medium sensor **114**, the third medium sensor **116**, the fourth medium sensor **117**, the fifth medium sensor **118**, and/or the sixth medium sensor **121**. Further, the light emitter and the light receiver may be provided to face each other with the medium conveyance path in between in the second medium sensor **114**, the third medium sensor **116**, the fourth medium sensor **117**, the fifth medium sensor **118**, and/or the sixth medium sensor **121**. Further, each of the second medium sensor **114**, the third medium sensor **116**, the fourth medium sensor **117**, the fifth medium sensor **118**, and/or the sixth medium sensor **121** may detect existence of a medium by, for example, a contact detection sensor to pass predetermined current when a medium is in contact or when a medium is not in contact. Further, each of the second medium sensor **114**, the third medium sensor **116**, the fourth medium sensor **117**, the fifth medium sensor **118**, and/or the sixth medium sensor **121** may include an arm, a light emitter and a light receiver, to detect the medium. The arm is provided to be able to move by the medium being in contact with the arm. The light emitter and the light receiver may be provided to face each other with the arm in between. Each sensor generates and outputs a medium signal, the signal value of which varies between a state in which a medium is present at the position of each sensor and a state in which a medium is not present, based on the intensity of light received by the light receiver.

(73) As illustrated in FIG. 4, the medium conveying apparatus **100** includes a first electromagnetic clutch **133**. The first electromagnetic clutch **133** is an example of an interruption mechanism and is provided on the shaft **112a** of the feed roller **112**, i.e., on a driving force transmission path from the first motor **131** to the feed roller **112**. The first electromagnetic clutch **133** is provided to be able to interrupt the driving force from the first motor **131** to the feed roller **112** in accordance with a control signal from the processing circuit.

(74) As illustrated in FIG. 5, the medium conveying apparatus **100** includes a second electromagnetic clutch **134**. The second electromagnetic clutch **134** is an example of an electromagnetic clutch and is provided on the shaft **113a** of the brake roller **113**, i.e., on a driving force transmission path from the second motor **132** being the driving source of the brake roller **113** to the brake roller **113**. The second electromagnetic clutch **134** is provided to be able to change the magnitude of torque exerted on the brake roller **113** in accordance with a control signal from the processing circuit.

(75) FIG. 6 is a block diagram illustrating a schematic configuration of an example of a medium conveying apparatus.

(76) In addition to the configuration described above, the medium conveying apparatus **100** further includes an interface device **135**, a storage device **140**, a processing circuit **150**, etc.

(77) For example, the interface device **135** includes an interface circuit conforming to a serial bus such as USB and transmits and receives an input image and various types of information by being electrically connected to an unillustrated information processing apparatus (such as a personal computer or a mobile information terminal). A communication device including an antenna transmitting and receiving wireless signals and a wireless communication interface circuit for transmitting and receiving signals through a wireless communication line in accordance with a predetermined communication protocol may be used in place of the interface device **135**. For example, the predetermined communication protocol is a wireless local area network (LAN). The

communication device may include a wired communication interface circuit for transmitting and receiving signals through a wired communication line in accordance with a communication protocol such as a wired LAN.

(78) The storage device **140** includes a memory device such as a random-access memory (RAM) or a read-only memory (ROM), a fixed disk device such as a hard disk, a portable storage device such as a flexible disk or an optical disk, etc. Further, a computer program, a database, a table, etc., that are used for various types of processing in the medium conveying apparatus **100** are stored in the storage device **140**. The computer programs may be installed on the storage device **140** from a computer-readable, non-transitory portable storage medium by using a well-known set-up program, etc. The portable storage medium is, for example, a compact disc read-only memory (CD-ROM) or a digital versatile disc read-only memory (DVD-ROM).

(79) The processing circuit **150** operates in accordance with a program previously stored in the storage device **140**. For example, the processing circuit is a central processing unit (CPU). Examples of the processing circuit **150** that may also be used include a digital signal processor (DSP), a large scale integration (LSI), an application specific integrated circuit (ASIC), and a field-programmable gate array (FPGA).

(80) The processing circuit **150** is connected to the operation device **105**, the display device **106**, the first medium sensor **111**, the second medium sensor **114**, the ultrasonic sensor **115**, the third medium sensor **116**, the fourth medium sensor **117**, the fifth medium sensor **118**, the sixth medium sensor **121**, the imaging device **122**, the first motor **131**, the second motor **132**, the first electromagnetic clutch **133**, the second electromagnetic clutch **134**, the interface device **135**, the storage device **140**, etc., and controls the components. The processing circuit **150** performs drive control of the first motor **131**, imaging control of the imaging device **122**, etc., based on each medium signal received from each sensor, acquires an input image from the imaging device **122**, and transmits the acquired image to the information processing apparatus through the interface device **135**.

(81) FIG. 7 is a diagram illustrating a schematic configuration of an example of a storage device and a processing circuit.

(82) As illustrated in FIG. 7, a control program **141**, a determination program **142**, etc., are stored in the storage device **140**. Each program is a functional module implemented by software operating on the processor. The processing circuit **150** reads each program stored in the storage device **140** and operates in accordance with the read program. Consequently, the processing circuit **150** functions as a control module **151**, and a determination module.

(83) FIG. 8 and FIG. 9 are flowcharts illustrating an operation example of medium reading processing in a medium conveying apparatus.

(84) The operation example of the medium reading process in the medium conveying apparatus **100** will be described below referring to the flowchart illustrated in FIG. 8 and FIG. 9. The operation flow described below is executed mainly by the processing circuit **150** in accordance with a program previously stored in the storage device **140** in cooperation with the components in the medium conveying apparatus **100**.

(85) First, the control module **151** waits until an instruction to read a medium is input by a user by using the operation device **105** or the information processing apparatus and an operation signal providing an instruction to read a medium is received from the operation device **105** or the interface device **132** (step **S101**).

(86) Next, the control module **151** acquires the first medium signal from the first medium sensor **111** and determines whether a medium is placed on the loading tray **103**, based on the acquired first medium signal (step **S102**). When a medium is not placed on the loading tray **103**, the control module **151** ends the series of steps.

(87) When a medium is placed on the loading tray **103**, the control module **151** sets surface moving speeds of the feed roller **112**, the brake roller **113**, the conveyance roller **119**, the first facing roller

120, the ejection roller **123**, and/or the second facing roller **124** (step **S103**). The surface moving speed is the speed at which a surface of each roller in contact with a medium moves. In other words, the surface moving speed of each of the feed roller **112**, the conveyance roller **119**, the first facing roller **120**, the ejection roller **123**, and/or the second facing roller **124** is the medium conveyance speed of each of the feed roller **112**, the conveyance roller **119**, and the ejection roller **123**. The surface moving speed of the brake roller **113** is the speed at which a surface of the brake roller **113** in contact with a medium moves in a direction opposite to the medium feeding direction. The surface moving speed of each roller may be hereinafter simply referred to as speed.

(88) The medium conveying apparatus **100** has a high-speed mode, a medium-speed mode, and a low-speed mode as conveyance modes of conveying a medium. The conveyance mode is set by a user by using the operation device **105** or the information processing apparatus, before medium reading process is executed.

(89) FIG. **10** to FIG. **12** are graphs for illustrating changes in the speeds of the feed roller **112**, the brake roller **113**, the conveyance roller **119**, the first facing roller **120**, the ejection roller **123**, and the second facing roller **124**. FIG. **10** illustrates changes in the speed of each roller in the high-speed mode, FIG. **11** illustrates changes in the speed of each roller in the medium-speed mode, and FIG. **12** illustrates changes in the speed of each roller in the low-speed mode.

(90) In FIG. **10** to FIG. **12**, graphs **G11**, **G21**, and **G31** illustrate changes in the speed of the feed roller **112**, graphs **G12**, **G22**, and **G32** illustrate changes in the speed of the brake roller **113**, and graphs **G13**, **G23**, and **G33** illustrate changes in the speed of the conveyance roller **119**. Since the speeds of the first facing roller **120**, the ejection roller **123**, and the second facing roller **124** change similarly to the speed of the conveyance roller **119**, changes in the speed of the conveyance roller **119** will be representatively described below. In each of the graphs **G11** to **G13**, **G21** to **G23**, and **G31** to **G33**, the horizontal axis indicates time, and the vertical axis indicates speed.

(91) A graph **G14** illustrates changes in the signal value of the fifth medium sensor **118**, and a graph **G15** illustrates changes in the signal value of the sixth medium sensor **121**. In each of the graphs **G14** and **G15**, the horizontal axis indicates time, and the vertical axis indicates the signal value. According to the present embodiment, when a medium does not exist at the position of each sensor, the signal value of a related signal takes on L, and when a medium exists at the position of each sensor, the signal value of a related signal takes on H.

(92) In FIG. **10** to FIG. **12**, a time **T1** denotes the time when medium feeding is started. As illustrated in FIG. **10** to FIG. **12**, the control module **151** sets the speed of the feed roller **112** to an initial speed **V1** at the start of medium feeding when the conveyance mode is set to any of the high-speed mode, the medium-speed mode, and the low-speed mode. When the conveyance mode is set to the high-speed mode, the control module **151** sets the speed of the brake roller **113** to an initial speed **U1** and sets the speed of the conveyance roller **119** to an initial speed **W1**. The initial speed **U1** of the brake roller **113** is set to a speed higher than $\frac{1}{2}$ of the initial speed **V1** of the feed roller **112**. Furthermore, the initial speed **U1** of the brake roller **113** may be set to a speed lower than the initial speed **V1** of the feed roller **112**.

(93) When the conveyance mode is set to the medium-speed mode, the control module **151** sets the speed of the brake roller **113** to a final speed **U3b** and sets the speed of the conveyance roller **119** to a final speed **W3b**. The final speed **U3b** of the brake roller **113** is set to a speed lower (slower) than a final speed **U3a** of the brake roller **113** in the high-speed mode described later. The final speed **W3b** of the conveyance roller **119** is set to a speed lower than a final speed **W3a** of the conveyance roller **119** in the high-speed mode described later. When the conveyance mode is set to the low-speed mode, the control module **151** sets the speed of the brake roller **113** to a final speed **U3c** and sets the speed of the conveyance roller **119** to a final speed **W3c**. The final speed **U3c** of the brake roller **113** is set to a speed lower than the final speed **U3b** of the brake roller **113** in the medium-speed mode. The final speed **W3c** of the conveyance roller **119** is set to a speed lower than the final speed **W3b** of the conveyance roller **119** in the medium-speed mode.

- (94) Next, the control module **151** drives the first motor **131** and the second motor **132**. Consequently, the control module **151** rotates the feed roller **112**, the brake roller **113**, the conveyance roller **119**, the first facing roller **120**, the ejection roller **123**, and/or the second facing roller **124** to feed and convey a medium (step **S104**).
- (95) The control module **151** controls the first motor **131** and the second motor **132** to rotate each roller at the set speed. As illustrated in FIG. **10** to FIG. **12**, each roller rotates at the set speed after an elapse of a predetermined through-up period from the start of drive of each motor at the time **T1**. When the control module **151** increases the speed of the roller, each roller rotates at the set speed after an elapse of the predetermined through-up period from the change of the speed of each motor. Further, when the control module **151** reduces the speed of the roller, each roller rotates at the set speed after an elapse of the predetermined through-down period from the change of the speed of each motor.
- (96) Next, the control module **151** waits until the front edge of the conveyed medium passes the position of the fifth medium sensor **118** (step **S105**). The control module **151** periodically acquires a fifth medium signal from the fifth medium sensor **118**, and determines that the front edge of the medium has passed the position of the fifth medium sensor **118** when the signal value of the fifth medium signal changes from a value indicating that a medium does not exist to a value indicating that a medium exists.
- (97) Next, the control module **151** changes the speeds of the feed roller **112** and the brake roller **113** (step **S106**).
- (98) In FIG. **10** to FIG. **12**, a time **T2** denotes the time when the signal value of the fifth medium signal has changed from L to H, i.e., when the front edge of the medium has passed the position of the fifth medium sensor **118**. As illustrated in FIG. **10** to FIG. **12**, the speed to which the feed roller **112** is changed when the front edge of the medium passes the position of the fifth medium sensor **118** varies depending on the conveyance mode.
- (99) When the conveyance mode is set to the high-speed mode, the control module **151** changes the speed of the feed roller **112** to a final speed **V3a**. The final speed **V3a** of the feed roller **112** is set to a speed higher than the initial speed **V1** of the feed roller **112** and lower than or equal to the final speed **W3a** of the conveyance roller **119** described later. The final speed **V3a** of the feed roller **112** may be set to a speed identical to the final speed **W3a** of the conveyance roller **119**. When the conveyance mode is set to the medium-speed mode, the control module **151** changes the speed of the feed roller **112** to a final speed **V3b**. The final speed **V3b** of the feed roller **112** in the medium-speed mode is set to a speed higher than the initial speed **V1** and lower than the final speed **V3a** in the high-speed mode. When the conveyance mode is set to the low-speed mode, the control module **151** changes the speed of the feed roller **112** to a final speed **V3c**. The final speed **V3c** of the feed roller **112** in the low-speed mode is set to a speed higher than the initial speed **V1** and lower than the final speed **V3b** in the medium-speed mode.
- (100) When the conveyance mode is set to the high-speed mode, the control module **151** changes the speed of the brake roller **113** to the final speed **U3a** and changes the speed of the conveyance roller **119** to the final speed **W3a**. The final speed **U3a** of the brake roller **113** is set to a speed higher than the initial speed **U1** of the brake roller **113**. The final speed **W3a** of the conveyance roller **119** is set to a speed higher than the initial speed **W1** of the conveyance roller **119**. When the conveyance mode is set to the medium-speed mode, the speeds of the brake roller **113** and the conveyance roller **119** are already set to the final speeds **U3b** and **W3b**, and therefore the control module **151** does not change the speeds of the brake roller **113** and the conveyance roller **119**. Similarly, when the conveyance mode is set to the low-speed mode, the speeds of the brake roller **113** and the conveyance roller **119** are already set to the final speeds **U3c** and **W3c**, and therefore the control module **151** does not change the speeds of the brake roller **113** and the conveyance roller **119**.
- (101) Next, the control module **151** waits until the front edge of the conveyed medium passes the

position of the conveyance roller **119** (step **S107**). The control module **151** periodically acquires a sixth medium signal from the sixth medium sensor **121** and determines that the front edge of the medium has passed the position of the sixth medium sensor **121** when the signal value of the sixth medium signal changes from a value indicating that a medium does not exist to a value indicating that a medium exists. The control module **151** determines that the front edge of the medium has passed the position of the conveyance roller **119** when the front edge of the medium passes the position of the sixth medium sensor **121**.

(102) Next, the control module **151** controls the first motor **131** to stop the feed roller **112** (step **S108**).

(103) In FIG. **10** to FIG. **12**, a time **T3** denotes the time when the signal value of the sixth medium signal has changed from L to H, i.e., when the front edge of the medium has passed the position of the sixth medium sensor **121**. As illustrated in FIG. **10** to FIG. **12**, after the front edge of the medium passes the position of the sixth medium sensor **121**, the control module **151** stops the feed roller **112** (change the speed to 0). Consequently, from this point, the medium is conveyed by the conveyance roller **119**, and the feed roller **112** is rotated by the medium being conveyed. By stopping the feed roller **112**, the control module **151** can suppress occurrence of jamming of the medium due to the medium being pushed by the feed roller **112** and being bent between the feed roller **112** and the conveyance roller **119**.

(104) Next, the control module **151** causes the imaging device **122** to start imaging of the medium (step **S109**).

(105) Next, the control module **151** waits until the rear edge of the conveyed medium passes the position of the fifth medium sensor **118** (step **S110**). The control module **151** periodically acquires the fifth medium signal from the fifth medium sensor **118**, and determines that the rear edge of the medium has passed the position of the fifth medium sensor **118** when the signal value of the fifth medium signal changes from a value indicating that a medium exists to a value indicating that a medium does not exist.

(106) Next, the control module **151** determines whether a medium remains on the loading tray **103**, based on the first medium signal received from the first medium sensor **111** (step **S111**).

(107) When a medium remains on the loading tray **103**, the control module **151** sets the speed of the feed roller **112** for feeding a succeeding medium (step **S112**).

(108) In FIG. **10** to FIG. **12**, a time **T4** denotes the time when the signal value of the fifth medium signal has changed from H to L, i.e., when the rear edge of the medium has passed the position of the fifth medium sensor **118**. As illustrated in FIG. **10** to FIG. **12**, the speed to which the feed roller **112** is set for feeding a succeeding medium when the rear edge of the medium passes the position of the fifth medium sensor **118** varies depending on the conveyance mode. When the conveyance mode is set to the high-speed mode or the medium-speed mode, the control module **151** sets the speed of the feed roller **112** to a first intermediate speed **V2a**. The first intermediate speed **V2a** is an example of a first speed and is set to a speed higher than the initial speed **V1** and lower than the final speed **V3a** and the final speed **V3b**. The first speed is the speed of the feed roller **112** in a first step when a second or subsequent medium is fed. The first intermediate speed **V2a** may be set to a speed almost identical to the final speed **V3c**. When the conveyance mode is set to the low-speed mode, the control module **151** sets the speed of the feed roller **112** to the final speed **V3c**.

(109) Next, by driving the first motor **131**, the control module **151** rotates the feed roller **112** to feed and convey the succeeding medium (step **S113**). The control module **151** controls the first motor **131** in such a way that the feed roller **112** rotates at a set speed.

(110) Next, the control module **151** waits until the front edge of the succeeding medium passes the position of the second medium sensor **114** (step **S114**). The control module **151** periodically acquires the second medium signal from the second medium sensor **114**, and determines that the front edge of the medium has passed the position of the second medium sensor **114** when the signal value of the second medium signal changes from a value indicating that a medium does not exist to

a value indicating that a medium exists.

(111) Next, the control module **151** changes the speed of the feed roller **112** (step **S115**).

(112) As illustrated in FIG. **10** to FIG. **12**, the speed to which the feed roller **112** is changed when the front edge of the succeeding medium passes the position of the second medium sensor **114** varies depending on the conveyance mode. When the conveyance mode is set to the high-speed mode, the control module **151** changes the speed of the feed roller **112** to a second intermediate speed **V2b**. The second intermediate speed **V2b** is an example of a second speed and is set to a speed higher than the first intermediate speed **V2a** and lower than the final speed **V3a**. The second speed is the speed of the feed roller **112** in a second step when a second or subsequent medium is fed. The second intermediate speed **V2b** may be set to a value almost identical to the final speed **V3b**. When the conveyance mode is set to the medium-speed mode, the control module **151** changes the speed of the feed roller **112** to the final speed **V3b**. When the conveyance mode is set to the low-speed mode, the control module **151** does not change the speed of the feed roller **112**.

(113) Next, the control module **151** waits until the rear edge of the preceding medium passes the imaging position of the imaging device **122** (step **S116**). The control module **151** periodically acquires the sixth medium signal from the sixth medium sensor **121**, and determines that the rear edge of the preceding medium has passed the position of the sixth medium sensor **121** when the signal value of the sixth medium signal changes from a value indicating that a medium exists to a value indicating that a medium does not exist. The control module **151** determines that the rear edge of the preceding medium has passed the imaging position when a first predetermined time elapses after the rear edge of the preceding medium passes the position of the sixth medium sensor **121**. The first predetermined time is set to a value acquired by adding a margin to the time required for a medium to move from the position of the sixth medium sensor **121** to the imaging position.

(114) Next, the control module **151** acquires an input image from the imaging device **122** and stores the acquired input image into the storage device **140**. The control module **151** outputs the input image stored in the storage device **140** by transmitting the image to the information processing apparatus through the interface device **135** and deletes the image from the storage device **140** (step **S117**).

(115) Next, the control module **151** waits until the front edge of the succeeding medium passes the position of the fifth medium sensor **118**, similarly to the processing in step **S105** (step **S118**).

(116) Next, the control module **151** changes the speed of the feed roller **112** (step **S119**).

(117) In FIG. **10** to FIG. **12**, a time **T5** denotes the time when the signal value of the fifth medium signal has changed from L to H, i.e., when the front edge of the succeeding medium has passed the position of the fifth medium sensor **118**. As illustrated in FIG. **10** to FIG. **12**, when the conveyance mode is set to the high-speed mode, the control module **151** changes the speed of the feed roller **112** to the final speed **V3a** when the front edge of the succeeding medium has passed the position of the fifth medium sensor **118**. When the conveyance mode is set to the medium-speed mode or the low-speed mode, the control module **151** does not change the speed of the feed roller **112**.

(118) Next, the control module **151** returns the processing to step **S107** and repeats the processing in and after step **S107** on the succeeding medium. In this case, the control module **151** waits until the front edge of the succeeding medium passes the position of the conveyance roller **119** (a time **T6** in FIG. **10**) in step **S107** and controls the first motor **131** to stop the feed roller **112** in step **S108**.

(119) When a medium does not remain on the loading tray **103** in step **S111**, the control module **151** waits until the rear edge of the conveyed medium passes the imaging position of the imaging device **122**, similarly to the processing in step **S116** (step **S120**).

(120) Next, the control module **151** acquires an input image from the imaging device **122** and outputs the acquired input image by transmitting the image to the information processing apparatus through the interface device **135** (step **S121**).

(121) Next, the control module **151** waits until the rear edge of the conveyed medium passes the position of the ejection roller **123** (step **S122**). The control module **151** determines that the rear

edge of the medium has passed the position of the ejection roller **123** when a second predetermined time elapses after the rear edge of the medium passes the position of the sixth medium sensor **121**. The second predetermined time is set to a value acquired by adding a margin to the time required for the medium to move from the position of the sixth medium sensor **121** to the position of the ejection roller **123**.

(122) Next, the control module **151** controls the second motor **132** to stop the brake roller **113**, the conveyance roller **119**, the first facing roller **120**, the ejection roller **123**, and/or the second facing roller **124** (step **S123**) and ends the series of steps.

(123) The control module **151** may execute the processing of step **S118** and after at step **S117** even when transmission of the input image is not completed. In that case, the control module **151** may temporarily stop feeding of the succeeding medium in step **S118** in a case of free space in the storage device **140** being less than a predetermined amount when the front edge of the succeeding medium passes the position of the fifth medium sensor **118**. The control module **151** controls the first motor **131** to stop the feed roller **112** until the free space in the storage device **140** becomes greater than or equal to the predetermined amount. Consequently, the medium conveying apparatus **100** can efficiently convey a medium while reliably transmitting an input image.

(124) The control module **151** may temporarily stop feeding of the succeeding medium in step **S118** in a case of the distance between the rear edge of the preceding medium and the front edge of the succeeding medium being short when the front edge of the succeeding medium passes the position of the fifth medium sensor **118**. For example, the control module **151** detects the time elapsed from when the rear edge of the preceding medium passes the position of the fifth medium sensor **118** or the sixth medium sensor **121** until the front edge of the succeeding medium passes the position of the fifth medium sensor **118**. When the measured time is less than a third predetermined time, the control module **151** controls the first motor **131** to stop the feed roller **112** until the third predetermined time elapses after the rear edge of the preceding medium passes the position of the fifth medium sensor **118** or the sixth medium sensor **121**. Consequently, the medium conveying apparatus **100** can efficiently convey a medium while suppressing occurrence of jamming of the medium.

(125) FIG. **13A** and FIG. **13B** are schematic diagrams for illustrating technical significance of feeding a medium in accordance with the medium reading process illustrated in FIG. **8** and FIG. **9**.

(126) FIG. **13A** and FIG. **13B** are schematic diagrams of the feed roller **112** and the brake roller **113** viewed from the side. FIG. **13A** illustrates a situation in which a first medium **M1** is fed in a state of a medium group **M** being placed on the loading tray **103**, and FIG. **13B** illustrates a situation in which a second or subsequent medium **M2** is fed in a state of the medium group **M** being placed on the loading tray **103**.

(127) As illustrated in the FIG. **13A**, when a certain amount of medium group **M** is placed on the loading tray **103**, the front edge of the medium group **M** is normally aligned by a user or an alignment member. Therefore, the front edge of the first medium **M1** placed at the bottom has not arrived at the nip region of the feed roller **112** and the brake roller **113**. In this case, the frictional force between media is increased by the weight of the medium group placed on the first medium **M1**. The medium group placed on the first medium **M1** tends to move to the downstream side with the first medium **M1** being fed by the frictional force and is pressed to the brake roller **113**. When the speed of the feed roller **112** at this time is excessively high, multi feed of media may occur due to the brake roller **113** lifted upward by the medium group placed on the first medium **M1** and the medium group entering between the feed roller **112** and the brake roller **113**.

(128) When the speed of the feed roller **112** at this time is excessively high, jamming of a medium may occur due to a lift of the front edge of the first medium **M1** before arriving at the nip region of the feed roller **112** and the brake roller **113**, and buckling of the medium **M1**.

(129) As described in steps **S103** to **S105** in FIG. **8**, when feeding the first medium of media placed on the loading tray **103**, the control module **151** controls the first motor **131** to rotate the feed roller

112 at a constant speed in a separation period from the start of medium feeding by the feed roller **112** to detection of the front edge of the medium by the fifth medium sensor **118**. The separation period when the first medium is fed is a period from the time **T1** to the time **T2** in FIG. **10** to FIG. **12**. In particular, when feeding the first medium, the control module **151** controls the first motor **131** to rotate the feed roller **112** at the initial speed **V1** lower than the first intermediate speed **V2a** at which a second or subsequent medium is fed. By reducing the speed of the feed roller **112** in the separation period of the first medium, the control module **151** can suppress occurrence of multi feed of media or jamming of a medium.

(130) As illustrated in the FIG. **13B**, when the first medium **M1** is fed, each second or subsequent medium **M2** is separated by the brake roller **113**, and the front edge of the medium **M2** is in a state of being in contact with the brake roller **113**. Similarly to the case of the first medium **M1** being fed, when the speed of the feed roller **112** is excessively high, multi feed of media or jamming of a medium may occur when each second or subsequent medium **M2** is fed as well. However, the possibility of occurrence of multi feed of medium or jamming of a medium when each second or subsequent medium **M2** is fed is less compared with the case of the first medium **M1** being fed. When the feeding speed of each second or subsequent medium **M2** in the separation period is excessively low, a considerable time is required for conveyance of all media to be completed. However, if the speed of the feed roller **112** at a standstill is sharply increased when feeding each second or subsequent medium **M2**, step out of the first motor **131** may occur.

(131) As described in steps **S112** to **S115** and **S118** in FIG. **9**, in a case where the conveyance mode is set to the high-speed mode, when feeding a second or subsequent medium, in the separation period, the control module **151** controls the first motor **131** to rotate the feed roller **112** at the first intermediate speed **V2a** and the second intermediate speed **V2b** that are higher than the initial speed **V1** at which the first medium is fed. The separation period when a second or subsequent medium is fed is a period from the time **T4** to the time **T5** in FIG. **10** to FIG. **12**. In particular, when feeding a second or subsequent medium of media placed on the loading tray **103**, the control module **151** controls the first motor **131** to rotate the feed roller **112** at the first intermediate speed **V2a** and then at the second intermediate speed **V2b**. Thus, the control module **151** increases the speed of the feed roller **112** in the separation period of the second or subsequent medium stepwise to speeds higher than the speed in the separation period of the first medium while reducing the speed in the separation period of the second or subsequent medium. Consequently, the control module **151** can reduce the processing time required for the medium reading process while suppressing occurrence of multi feed of media or jamming of a medium, and step out of the first motor **131**; and higher feeding performance (reduction in occurrence of an abnormality) and higher processing performance (reduction in the conveyance time) can be simultaneously achieved.

(132) FIG. **14** is a schematic diagram for illustrating technical significance of feeding a medium in accordance with the medium reading process illustrated in FIG. **8** and FIG. **9**. FIG. **14** is a schematic diagram of the lower guide **101a** of the lower housing **101** viewed from above.

(133) FIG. **14** illustrates a state in which the front edge of a medium **M3** conveyed in a tilted manner is in contact with either one of the two conveyance rollers **119** (first facing rollers **120**). When the speed of the conveyance rollers **119** is almost identical to the speed of the feed rollers **112**, the medium **M3** is subsequently conveyed to the downstream side while maintaining the current tilt. However, when the speed of the conveyance rollers **119** is higher than the speed of the feed rollers **112**, the medium **M3** is subsequently pulled by the conveyance roller **119** on the contacting side and rotates in a direction of an arrow **A9**. Accordingly, jamming of the medium may occur due to an increase in the tilt of the medium **M3** and collision of the medium **M3** with a side wall of the conveyance path. As the difference between the speed of the conveyance rollers **119** and the speed of the feed rollers **112** increases, the possibility of occurrence of jamming of a medium increases.

(134) As described above, the control module **151** may temporarily stop medium feeding in a case

of free space in the storage device **140** being insufficient or the distance between media being short when the front edge of the medium passes the position of the fifth medium sensor **118**. However, when the control module **151** temporarily stops medium feeding, it takes a certain through-up period after medium feeding is subsequently resumed until the speed of the feed roller **112** arrives at a set speed. When the speed of the feed roller **112** is not sufficiently increased at the point in time when the front edge of the medium arrives at the position of the conveyance roller **119**, a skew of the medium may occur. Therefore, when temporarily stopping medium feeding, the control module **151** needs to stop the medium at a position on a sufficiently upstream side of the conveyance roller **119**. However, it takes a certain through-down period after the control module **151** stops the first motor **131** until rotation of the feed roller **112** completely stops. For example, the aforementioned issue can be avoided by sufficiently increasing the distance between the feed roller **112** and the conveyance roller **119**; however, in that case, the size of the entire device increases.

(135) As described in steps **S106** and **S119** in FIG. **8** and FIG. **9**, in the high-speed mode, the control module **151** controls the first motor **131** to rotate the feed roller **112** at the final speed **V3a** higher than the second intermediate speed **V2b** after the fifth medium sensor **118** detects the front edge of a medium. Therefore, the speed of the feed roller **112** is set to the second intermediate speed **V2b** lower than the final speed **V3a** at the point in time when the front edge of the medium passes the position of the fifth medium sensor **118**, and the control module **151** can completely stop rotation of the feed roller **112** in a sufficiently short period. In the medium-speed mode and the low-speed mode, the speed of the feed roller **112** is set to the final speed **V3b** and the final speed **V3c**, respectively, at the point in time when the front edge of the medium passes the position of the fifth medium sensor **118**. However, since the final speed **V3b** and the final speed **V3c** are lower than the final speed **V3a**, the control module **151** can completely stop rotation of the feed roller **112** in a sufficiently short period.

(136) Accordingly, when temporarily stopping medium feeding, the control module **151** can stop the medium at a position on a sufficiently upstream side of the conveyance roller **119**. Consequently, when resuming medium feeding, the control module **151** can sufficiently increase the speed of the feed roller **112** before the front edge of the medium arrives at the position of the conveyance roller **119** and can suppress an increase in the tilt of the medium and suppress occurrence of jamming of the medium.

(137) As described in steps **S114** and **S115** in FIG. **9**, in a case of feeding a second or subsequent medium, when the second medium sensor **114** detects the front edge of the medium in the separation period, the control module **151** controls the first motor **131** to rotate the feed roller **112** at the second intermediate speed **V2b**. After the front edge of the medium passes the position of the second medium sensor **114**, i.e., after the front edge passes the nip region of the feed roller **112** and the brake roller **113**, a lift of the front edge of the medium due to collision with the brake roller **113** does not occur. Therefore, the possibility of occurrence of jamming of the medium due to buckling of the medium is low. The control module **151** determines that the front edge of the medium has passed the nip region of the feed roller **112** and the brake roller **113** when the second medium sensor **114** detects the front edge of the medium and increases the speed of the feed roller **112**. Consequently, the control module **151** can reduce the medium feeding time while suppressing occurrence of jamming of the medium.

(138) In particular, by a decrease in the period in which the feed roller **112** rotates at the first intermediate speed **V2a** and an increase in the period in which the feed roller **112** rotates at the second intermediate speed **V2b**, the medium feeding speed can be increased as a whole even when the second intermediate speed **V2b** is low. In other words, when temporarily stopping medium feeding as described above, the control module **151** can decrease the second intermediate speed **V2b**, and can stop the medium at a position on a sufficiently upstream side of the conveyance roller **119**. In steps **S114** and **S115** in FIG. **9**, the control module **151** may change the speed of the feed roller **112** when a fourth predetermined time elapses after the start of medium feeding instead of

when the front edge of the medium passes the position of the second medium sensor **114**. The fourth predetermined time is set to the time required for the front edge of the medium to pass the nip region of the feed roller **112** and the brake roller **113** after the start of medium feeding through a previously performed experiment. In that case, the second medium sensor **114** may be omitted. When the second medium sensor **114** is omitted, the fifth medium sensor **118** may be located at the position of the second medium sensor **114**.

(139) The control module **151** may increase the amount of current supplied to the first motor **131** and/or the second motor **132** in the through-up period of the first motor **131** and/or the second motor **132**. Consequently, step out of each motor is less likely to occur even when the speed of the motor is increased in a short period, and the distance required for through-up can be reduced. Further, by limiting the period in which the amount of current is increased to the through-up period, the medium conveying apparatus **100** can suppress an increase in the amount of power consumption as a whole.

(140) As illustrated in FIG. **13B**, a medium entering the nip region of the feed roller **112** and the brake roller **113** is pushed back by the brake roller **113** rotating in a direction opposite to the medium feeding direction. As the ratio of the speed of the brake roller **113** to the speed of the feed roller **112** increases, the medium is more satisfactorily pushed back. In particular, as illustrated in FIG. **13A**, when feeding of the first medium **M1** starts, the front edge of each medium placed above the first medium **M1** has not arrived at the nip region of the feed roller **112** and the brake roller **113**. Therefore, when feeding of the first medium **M1** starts, each medium enters the nip region of the feed roller **112** and the brake roller **113** with force. Accordingly, when the first medium **M1** is fed, the ratio of the speed of the brake roller **113** to the speed of the feed roller **112** needs to be set to a somewhat large value such that other media are satisfactorily returned to the upstream side.

(141) As illustrated in FIG. **13B**, when a second or subsequent medium **M2** is fed, the front edge of each medium **M2** is already in contact with the brake roller **113**. Accordingly, when a second or subsequent medium **M2** is fed, a medium other than the medium being fed is satisfactorily pushed back even when the ratio of the speed of the brake roller **113** to the speed of the feed roller **112** is set to a somewhat small value.

(142) As described above, the brake roller **113** is driven by the same second motor **132** as the conveyance roller **119** and the ejection roller **123**. Consequently, the medium conveying apparatus **100** can achieve reduction in the apparatus cost and the apparatus size by reducing the number of motors. However, when the speed of the brake roller **113** is reduced, the speeds of the conveyance roller **119** and the ejection roller **123** are also reduced, and medium processing performance is deteriorated.

(143) As the speed of the brake roller **113** increases, vibrations of the brake roller **113** increase and a loud vibration sound (so-called chattering sound) is generated. When feeding of a second or subsequent medium is started, the medium is in contact with the brake roller **113** and vibrations are suppressed; however, when feeding of the first medium **M1** is started, the medium is not in contact with the brake roller **113** and vibrations are not suppressed. Accordingly, it is desirable to reduce the speed of the brake roller **113** particularly when feeding of the first medium **M1** is started.

(144) As described in steps **S103** and **S106** in FIG. **8**, the control module **151** sets the rotation speed of the brake roller **113** in the separation period when the first medium of media placed on the loading tray **103** is fed to a speed lower than the rotation speed of the brake roller **113** in the separation period when a second or subsequent medium is fed. Consequently, the control module **151** can suppress vibrations of the brake roller **113** when the first medium is fed and suppress occurrence of a vibration sound. In the separation period, when the first medium is fed, the conveyance roller **119** has not conveyed the medium yet, and the medium conveyance time does not increase even when the rotation speed of the brake roller **113** is reduced. By increasing the rotation speed of the brake roller **113** in the separation period when a second or subsequent medium is fed, the control module **151** can keep the rotation speed of the conveyance roller **119** high.

Accordingly, the medium conveying apparatus **100** can improve processing performance (reduction in the conveyance time) while suppressing an increase in the apparatus cost and the apparatus size. (145) As described in step **S103** in FIG. **8**, the control module **151** sets the speed of the feed roller **112** to the sufficiently low initial speed **V1** and sets the initial speed **U1** of the brake roller **113** to a speed higher than $\frac{1}{2}$ of the initial speed **V1** of the feed roller **112** at the start of medium feeding. In other words, the control module **151** sets the rotation speed of the brake roller **113** in such a way that the speed of the brake roller **113** in the separation period when the first medium of media placed on the loading tray **103** is fed is higher than $\frac{1}{2}$ of the speed of the feed roller **112** in the separation period when the first medium is fed. Consequently, the brake roller **113** can satisfactorily push back a medium entering the nip region of the feed roller **112** and the brake roller **113** at the time of medium feeding.

(146) As described in detail above, the medium conveying apparatus **100** increases the rotation speed of the feed roller **112** stepwise in a period from the start of medium feeding until the front edge of the medium passes the feed roller **112**. Consequently, the medium conveying apparatus **100** can suitably control stoppage and resumption of medium feeding, for example, when free space in the storage device **140** is insufficient or when the distance between successively fed media is short, while reducing the medium feeding time. Accordingly, the medium conveying apparatus **100** can control medium feeding better.

(147) In particular, the medium conveying apparatus **100** can control medium feeding better by devising control method of a motor without using a motor or a medium sensor with a special structure. Accordingly, the medium conveying apparatus **100** controls medium feeding better while suppressing an increase in the apparatus cost and the apparatus size.

(148) By rotating the feed roller **112** at a low speed during medium separation and rotating the feed roller **112** at a high speed when medium separation is not performed, the medium conveying apparatus **100** can convey a medium in a short period while satisfactorily separating the medium.

(149) FIG. **15** is a schematic diagram for illustrating driving sources of a feed roller **112**, a brake roller **113**, a conveyance roller **219**, a first facing roller **220**, an ejection roller **223**, and/or a second facing roller **224** in a medium conveying apparatus according to another embodiment.

(150) As illustrated in FIG. **15**, the medium conveying apparatus according to the present embodiment includes the conveyance roller **219**, the first facing roller **220**, the ejection roller **223**, and the second facing roller **224** in place of the conveyance roller **119**, the first facing roller **120**, the ejection roller **123**, and the second facing roller **124**. Further, the medium conveying apparatus includes a second motor **232** in place of the second motor **132**.

(151) The conveyance roller **219** is provided in a lower housing **101**, and the first facing roller **220** is provided on the upper side of the conveyance roller **219** in an upper housing **102**. The ejection roller **223** is provided in the lower housing **101**, and the second facing roller **224** is provided on the upper side of the ejection roller **223** in the upper housing **102**.

(152) The second motor **232** is provided in the lower housing **101** separately from a first motor **131**, is connected to the conveyance roller **219**, the ejection roller **223**, and the brake roller **113** through a second transmission mechanism **232a**, and drives the conveyance roller **219**, the ejection roller **223**, and the brake roller **113**. The second motor **232** generates a driving force for driving the conveyance roller **219**, the ejection roller **223**, and the brake roller **113** in accordance with a control signal from a processing circuit **150**. The second transmission mechanism **232a** includes one or a plurality of pulleys, belts, gears, etc., provided between the second motor **232**, and a shaft **219a** of the conveyance roller **219**, a shaft **223a** of the ejection roller **223**, and a shaft **113a** of the brake roller **113**. In particular, one or a plurality of gears for changing the rotation direction and the rotation speed of each roller are provided between the shaft **219a** of the conveyance roller **219** and/or the shaft **223a** of the ejection roller **223**, and the shaft **113a** of the brake roller **113**. The gears are located outside a medium conveyance path in a width direction **A8** to transmit the driving force with the medium conveyance path in between. The second transmission mechanism **232a**

transmits the driving force generated by the second motor **232** to the conveyance roller **219**, the ejection roller **223**, and the brake roller **113**. Consequently, the second motor **232** causes the conveyance roller **219**, the ejection roller **223**, and the brake roller **113** to feed, convey, and eject a medium by rotating the conveyance roller **219**, the ejection roller **223**, and the brake roller **113**. The second motor **232** is an example of a driving source of the brake roller **113**.

(153) The first facing roller **220** is a driven roller driven to rotate by the conveyance roller **219**, and the second facing roller **224** is a driven roller driven to rotate by the ejection roller **223**. The first facing roller **220** and/or the second facing roller **224** may be provided to be driven by the driving force from the second motor **232**. In that case, one or a plurality of gears are further provided between the shaft **219a** of the conveyance roller **219** and a shaft **220a** of the first facing roller **220**, and/or between the shaft **223a** of the ejection roller **223** and a shaft **224a** of the second facing roller **224**. The second transmission mechanism **232a** further transmits the driving force generated by the second motor **232** to the first facing roller **220** and/or the second facing roller **224**.

(154) As described in detail above, the medium conveying apparatus can control medium feeding better even when driving the conveyance roller **219** and the brake roller **113** by the same second motor **232** while providing the conveyance roller **219** in the lower housing **101**.

(155) FIG. **16** is a flowchart illustrating an example of part of operation of medium reading process in a medium conveying apparatus according to another embodiment.

(156) The flowchart illustrated in FIG. **16** is executed in place of the flowchart illustrated in FIG. **9**. Processing in steps **S211** to **S213** and **S216** to **S223** in FIG. **16** is similar to the processing in steps **S111** to **S113** and **S116** to **S123** in FIG. **9**; and therefore description thereof is omitted, and only steps **S214** and **S215** will be described below.

(157) After a control module **151** drives a feed roller **112** in step **S213**, a determination module **152** determines whether a skew of the conveyed medium has occurred (step **S214**). The determination module **152** periodically acquires the third medium signal and the fourth medium signal from the third medium sensor **116** and the fourth medium sensor **117** and determines whether a skew of the medium has occurred, based on the acquired third medium signal and the acquired fourth medium signal. The determination module **152** determines that the front edge of the medium has passed the position of the third medium sensor **116** when the signal value of the third medium signal changes from a value indicating that a medium does not exist to a value indicating that a medium exists. Further, the determination module **152** determines that the front edge of the medium has passed the position of the fourth medium sensor **117** when the signal value of the fourth medium signal changes from a value indicating that a medium does not exist to a value indicating that a medium exists.

(158) The determination module **152** determines that a skew of the medium has occurred when the front edge of the medium does not pass one of the positions of the third medium sensor **116** and the fourth medium sensor **117** within a fifth predetermined time after passing the other position. The fifth predetermined time is set to, through a previously performed experiment, a value between the difference between passing times of the positions of the sensors when jamming of a medium has not occur and the difference between passing times of the positions of the sensors when jamming of a medium occurs. The determination module **152** determines that a skew of the medium has not occurred when the front edge of the medium passes one of the positions of the third medium sensor **116** and the fourth medium sensor **117** within the fifth predetermined time after passing the other position.

(159) Next, the control module **151** changes the speed of the feed roller **112** (step **S215**). Similarly to the processing in step **S115**, the control module **151** changes the speed of the feed roller **112** to a final speed **V3b** when the conveyance mode is set to a medium-speed mode and does not change the speed of the feed roller **112** when the conveyance mode is set to a low-speed mode. When the conveyance mode is set to a high-speed mode, the control module **151** changes the speed of the feed roller **112**, based on the determination result of a skew of the medium. The control module **151**

may change the speed of the feed roller **112** based on the difference between passing time of the position of the third medium sensor **116** and passing time of the position of the fourth medium sensor **117** without determining a skew of the medium has occurred. For example, when the difference is more than a predetermined value, the control module **151** sets a speed of the feed roller **112** to a speed lower than a speed of the feed roller **112** when the difference is less than or equal to the predetermined value.

(160) FIG. **17** is a graph for illustrating changes in the speed of the feed roller **112** in the high-speed mode.

(161) In FIG. **17**, a graph **G41** illustrates changes in the speed of the feed roller **112**. In the graph **G41**, the horizontal axis indicates time, and the vertical axis indicates speed. The speeds of a brake roller **113** and a conveyance roller **119** change similarly to the graphs **G12** and **G13** illustrated in FIG. **10**. Graphs **G14** and **G15** illustrate changes in the signal values of a fifth medium sensor **118** and a sixth medium sensor **121** and change similarly to the graphs **G14** and **G15** illustrated in FIG. **10**. Times **T1** to **T6** indicate times similar to the times **T1** to **T6** indicated in FIG. **10**.

(162) When a skew of a medium occurs, the control module **151** changes the speed of the feed roller **112** to a second intermediate speed **V2b**, as illustrated in the graph **G11** illustrated in FIG. **10**. When a skew of a medium does not occur, the control module **151** changes the speed of the feed roller **112** to a final speed **V3a**, as the graph **G41** illustrated in FIG. **17**. In this case, the final speed **V3a** is an example of a second speed. The final speed **V3a** is set to a speed identical to the speed of the feed roller **112**. In other words, when the determination module **152** determines that a skew of a medium has occurred, the control module **151** sets the second intermediate speed **V2b** as the second speed in such a way that the speed of the feed roller **112** is lower than the speed of the conveyance roller **119**. When the determination module **152** does not determine that a skew of a medium has occurred, the control module **151** sets the final speed **V3a** as the second speed in such a way that the speed of the feed roller **112** is identical to the speed of the conveyance roller **119**.

(163) As described above, when free space in a storage device **140** is insufficient or when the distance between media is short, the control module **151** may temporarily stop medium feeding. However, when a medium is tilted and the speed of the conveyance roller **119** is higher than the speed of the feed roller **112**, jamming of the medium may occur due to an increase in the tilt of the medium and collision of the medium with a side wall of a conveyance path. By setting the speed of the feed roller **112** to the second intermediate speed **V2b** when a skew of a medium occurs, the control module **151** can stop the medium at a position on a sufficiently upstream side of the conveyance roller **119** when temporarily stopping medium feeding. Consequently, the control module **151** can sufficiently increase the speed of the feed roller **112** before the front edge of the medium arrives at the position of the conveyance roller **119** when resuming medium feeding and can suppress increase in the tilt of the medium and suppress occurrence of jamming of the medium.

(164) When the medium is not tilted, the possibility of occurrence of jamming of the medium due to an increase in the tilt of the medium is low even when the speed of the conveyance roller **119** is higher than the speed of the feed roller **112**. By setting the speed of the feed roller **112** to the final speed **V3a** when a skew of the medium does not occur, the control module **151** can reduce the medium feeding time. Accordingly, the control module **151** can reduce the medium conveyance time while suppressing occurrence of jamming of the medium.

(165) As described in detail above, the medium conveying apparatus can control medium feeding better when the speed of the feed roller **112** is changed depending on whether a skew of a medium has occurred as well.

(166) FIG. **18** is a flowchart illustrating an example of part of operation of medium reading process in a medium conveying apparatus according to yet another embodiment.

(167) The flowchart illustrated in FIG. **18** is executed in place of the flowchart illustrated in FIG. **8**. Processing in steps **S301** to **S306** and **S308** to **S311** in FIG. **18** is similar to the processing in steps **S101** to **S106** and **S107** to **S110** in FIG. **8**; and therefore, description thereof is omitted, and only

steps **S307** and **S312** will be described below.

(168) After waiting until the front edge of the medium passes the position of a fifth medium sensor **118** in step **S305** or **S118**, i.e., after the front edge of the medium passes the position of an ultrasonic sensor **115**, a control module **151** detects the thickness of a conveyed medium (step **S307**). The control module **151** detects the thickness of the medium, based on the ultrasonic signal received from the ultrasonic sensor **115**. An ultrasonic wave transmitted by the ultrasonic transmitter **115a** and passing through a medium is attenuated by the medium, and as the thickness of the medium increases, the amount of attenuation of the ultrasonic wave increases. The medium conveying apparatus **100** previously stores a table defining a relation between the magnitude of the ultrasonic wave received by the ultrasonic receiver **115b**, i.e., the signal value of the ultrasonic signal, and the thickness of a medium into a storage device **140**. The control module **151** specifies the thickness of a medium related to the signal value of the received ultrasonic signal with reference to the table stored in the storage device **140**.

(169) The control module **151** may further determine whether multi feed of media has occurred, based on the ultrasonic signal received from the ultrasonic sensor **115**. When a plurality of media are conveyed in a stacked manner, an ultrasonic wave passing through the media is attenuated by an air layer between media conveyed in a stacked manner. Accordingly, the control module **151** can determine whether multi feed of media has occurred, based on whether the signal value of the ultrasonic signal is less than or equal to a multi feed threshold value. The multi feed threshold value is set to a value between the signal value of the ultrasonic signal when one sheet of paper is conveyed and the signal value of the ultrasonic signal when two sheets of paper are conveyed. When determining that multi feed of media has occurred, the control module **151** stops conveyance and ejection of a medium by stopping a first motor **131** and a second motor **132**. The control module **151** may stop the medium reading process after ejecting a currently conveyed medium. The control module **151** may control each roller to, by driving each motor, reversely feed media remaining in a conveyance path, temporarily return the media to a loading tray **103**, and refeed (separate) the media. Consequently, a user does not need to replace the media onto the loading tray **103** and refeed the media, and the control module **151** can improve user convenience. Further, the control module **151** may notify a user of information indicating that multi feed of media has occurred by displaying the information on a display device **106** or transmitting the information to an information processing apparatus through an interface device **135**.

(170) The control module **151** may detect the thickness of a medium by using a thickness sensor other than the ultrasonic sensor **115**. The thickness sensor is located at a position where the ultrasonic sensor **115** is located. The thickness sensor may be located at any position on a medium conveyance path. For example, the thickness sensor is a reflected light sensor including a pair of a light emitter and a light receiver that are provided on one side of the medium conveyance path and a pair of a light emitter and a light receiver that are provided on the other side. The reflected light sensor detects the distance between each pair and each surface of a medium based on the time elapsed after one pair emits light onto one surface of the medium before the pair receives reflected light and the time elapsed after the other pair emits light onto the other surface of the medium before the pair receives reflected light. The reflected light sensor generates a thickness signal indicating a subtracted value acquired by subtracting the detected distances from the distance between the two pairs as the thickness. The medium conveying apparatus **100** previously stores a table defining a relation between the signal value of the thickness signal and the thickness of a medium into the storage device **140**. The control module **151** specifies the thickness of a medium related to the signal value of the received thickness signal with reference to the table stored in the storage device **140**. The thickness sensor is not limited to a sensor using light, and any other sensor that can detect the thickness of a medium, such as a pressure sensor or a thickness sensor using a contact segment, may be used as the thickness sensor.

(171) After waiting until the rear edge of the medium passes the position of the fifth medium sensor

118 in step **S311**, the control module **151** detects the size of the conveyed medium (step **S312**). For example, the control module **151** detects the length of the medium in a medium conveying direction **A1** as the size of the medium. The control module **151** detects the length of the medium in the medium conveying direction **A1**, based on a fifth medium signal received from the fifth medium sensor **118**. The control module **151** calculates the movement distance of the medium by a feed roller **112** by driving the first motor **131** in a period from detection of the front edge of the medium to detection of the rear edge of the medium by the fifth medium sensor **118** as the length of the medium in the medium conveying direction **A1**. In other words, the control module **151** calculates a value acquired by multiplying the time period from detection of the front edge of the medium to detection of the rear edge of the medium by the fifth medium sensor **118** by the conveyance speed of the medium as the length of the medium in the medium conveying direction **A1**.

(172) The control module **151** may detect the length of the medium in a width direction **A8** as the size of the medium. In that case, a number of fifth medium sensors **118** are spaced in the width direction **A8** in the medium conveying apparatus **100**, and placement intervals of the fifth medium sensors **118** are previously stored. The control module **151** detects the length of a conveyed medium in the width direction **A8**, based on the distance between the outermost fifth medium sensors **118** detecting the medium.

(173) The control module **151** may detect the length of a medium in the medium conveying direction **A1** or the length of the medium in the width direction **A8**, based on an input image generated by an imaging device **122** when imaging of the medium is completed. In that case, the control module **151** detects an edge of the medium from the input image by using a known image processing technology and detects the length of the medium in the medium conveying direction **A1** or the width direction **A8**, based on the distance between the upper edge and the lower edge of the medium or the distance between the left edge and the right edge.

(174) When the flowchart illustrated in FIG. **18** is executed, the control module **151** changes the speed of the feed roller **112**, based on the length of the medium, in step **S115** and/or **S119** in the flowchart illustrated in FIG. **9**.

(175) FIG. **19** is a graph for illustrating changes in the speed of the feed roller **112** in a high-speed mode.

(176) In FIG. **19**, graphs **G51** and **G52** illustrate examples of changes in the speed of the feed roller **112**. In each of the graphs **G51** and **G52**, the horizontal axis indicates time, and the vertical axis indicates speed. The speeds of a brake roller **113** and a conveyance roller **119** change similarly to those illustrated in the graphs **G12** and **G13** in FIG. **10**. Graphs **G14** and **G15** illustrate changes in the signal values of the fifth medium sensor **118** and a sixth medium sensor **121** and change similarly to the graphs **G14** and **G15** illustrated in FIG. **10**. Times **T1** to **T6** indicate times similar to the times **T1** to **T6** indicated in FIG. **10**.

(177) In step **S115**, when the size of the medium is less than or equal to a first size threshold value and greater than a second size threshold value, the control module **151** changes the speed of the feed roller **112** to the second intermediate speed **V2b** as illustrated in the graph **G11** illustrated in FIG. **10**. For example, the first size threshold value is set to a value between the **A4** longitudinal size and the **A3** longitudinal size when the size of the medium is the length of the medium in the medium conveying direction **A1** and is set to a value between the **A4** lateral size and the **A3** lateral size when the size of the medium is the length of the medium in the width direction **A8**. The second size threshold value is set to a value less than the first size threshold value. For example, the second size threshold value is set to a value between the **A5** longitudinal size and the **A4** longitudinal size when the size of the medium is the length of the medium in the medium conveying direction **A1** and is set to a value between the **A5** lateral size and the **A4** lateral size when the size of the medium is the length of the medium in the width direction **A8**. The first size threshold value and the second size threshold value are examples of a size threshold value.

(178) When the size of the medium is greater than the first size threshold value, the control module **151** changes the speed of the feed roller **112** to a second intermediate speed $V2c$ as illustrated in the graph G51. The second intermediate speed $V2c$ is set to a speed higher than the second intermediate speed $V2b$ when the size of the medium is less than or equal to the first size threshold value and greater than the second size threshold value, and lower than the final speed $V3a$. When the size of the medium is less than or equal to the second size threshold value, the control module **151** changes the speed of the feed roller **112** to a second intermediate speed $V2d$ as illustrated in the graph G52. The second intermediate speed $V2d$ is set to a speed lower than the second intermediate speed $V2b$ when the size of the medium is less than or equal to the first size threshold value and greater than the second size threshold value, and higher than the first intermediate speed $V2a$.

(179) In step S119, when the size of the medium is less than or equal to the first size threshold value and greater than the second size threshold value, the control module **151** changes the speed of the feed roller **112** to the final speed $V3a$ as illustrated in the graph G11 illustrated in FIG. 10. When the size of the medium is greater than the first size threshold value, the control module **151** changes the speed of the feed roller **112** to a final speed $V3d$ as illustrated in the graph G51. The final speed $V3d$ is set to a speed higher than the final speed $V3a$ at which the feed roller **112** is set when the size of the medium is less than or equal to the first size threshold value and greater than the second size threshold value. When changing the speed of the feed roller **112** to the final speed $V3d$, the control module **151** may change the speed of the feed roller **112** to the final speed $V3a$ before the front edge of the succeeding medium passes the position of the fifth medium sensor **118**, as illustrated in a dotted line D1 in the graph G51. In other words, the control module **151** decelerates the speed of the feed roller **112** to the final speed $V3a$ before the front edge of the succeeding medium passes the position of the conveyance roller **119**. Consequently, the control module **151** can suppress occurrence of jamming of the medium due to the medium being pushed by the feed roller **112** and being bent between the feed roller **112** and the conveyance roller **119**. When the size of the medium is less than or equal to the second size threshold value, the control module **151** changes the speed of the feed roller **112** to the second intermediate speed $V2b$ as illustrated in the graph G52.

(180) When the size of the medium is greater than the first size threshold value, the control module **151** may change the speed of the feed roller **112** to a speed similar to the speed when the size of the medium is less than or equal to the first size threshold value and greater than the second size threshold value. Alternatively, when the size of the medium is less than or equal to the second size threshold value, the control module **151** may change the speed of the feed roller **112** to a speed similar to the speed when the size of the medium is less than or equal to the first size threshold value and greater than the second size threshold value.

(181) Thus, when the size of a preceding medium is small, the control module **151** lowers the speed of the feed roller **112** after the front edge of a succeeding medium passes a nip region of the feed roller **112** and the brake roller **113** than the speed when the size of the preceding medium is large. In other words, when the size of the preceding medium is less than or equal to a size threshold value, the control module **151** sets a second speed to a speed lower than the second speed when the size of the preceding medium is greater than the size threshold value.

(182) In general, as the length of a medium decreases, the distance between successively conveyed media tends to decrease. The control module **151** can suppress occurrence of collision between successively conveyed media by lowering the speed of the feed roller **112** when the size of the preceding medium is small than the speed of the feed roller **112** when the size of the preceding medium is large.

(183) When the flowchart illustrated in FIG. 18 is executed, the control module **151** may change the speed of the feed roller **112** in step S115 and/or S119 in the flowchart illustrated in FIG. 9 based on the thickness of the medium.

(184) In step S115, when the thickness of the medium is less than or equal to a first thickness

threshold value and greater than a second thickness threshold value, the control module **151** changes the speed of the feed roller **112** to the second intermediate speed $V2b$ as the graph **G11** illustrated in FIG. **10**. For example, the first thickness threshold value is set to a value between the thickness of plain paper copier (PPC) paper and the thickness of a common business card. The second thickness threshold value is set to a value less than the first thickness threshold value. For example, the second thickness threshold value is set to a value between the thickness of common thin paper and the thickness of PPC paper. The first thickness threshold value and the second thickness threshold value are examples of a thickness threshold value.

(185) When the thickness of the medium is greater than the first thickness threshold value, the control module **151** changes the speed of the feed roller **112** to the second intermediate speed $V2c$ as the graph **G51**. When the thickness of the medium is less than or equal to the second thickness threshold value, the control module **151** changes the speed of the feed roller **112** to the second intermediate speed $V2d$ as the graph **G52**.

(186) In step **S119**, when the thickness of the medium is less than or equal to the first thickness threshold value and greater than the second thickness threshold value, the control module **151** changes the speed of the feed roller **112** to the final speed $V3a$ as the graph **G11** illustrated in FIG. **10**. When the thickness of the medium is greater than the first thickness threshold value, the control module **151** changes the speed of the feed roller **112** to the final speed $V3d$ as the graph **G51**. When the size of the medium is less than or equal to the second thickness threshold value, the control module **151** changes the speed of the feed roller **112** to the second intermediate speed $V2b$ as the graph **G52**.

(187) When the thickness of the medium is greater than the first thickness threshold value, the control module **151** may change the speed of the feed roller **112** to a speed similar to the speed when the thickness of the medium is less than or equal to the first thickness threshold value and greater than the second thickness threshold value. Alternatively, when the thickness of the medium is less than or equal to the second thickness threshold value, the control module **151** may change the speed of the feed roller **112** to a speed similar to the speed when the thickness of the medium is less than or equal to the first thickness threshold value and greater than the second thickness threshold value.

(188) Thus, when a preceding medium is thin, the control module **151** lowers the speed of the feed roller **112** after the front edge of a succeeding medium passes the nip region of the feed roller **112** and the brake roller **113** than the speed when the preceding medium is thick. In other words, when the thickness of the preceding medium is less than or equal to a thickness threshold value, the control module **151** sets the second speed to a speed lower than the second speed when the thickness of the preceding medium is greater than the thickness threshold value.

(189) In general, as the thickness of a medium decreases, the possibility of occurrence of jamming of the medium tends to increase. The control module **151** can suppress the occurrence of jamming of the medium by lowering the speed of the feed roller **112** when a preceding medium is thin than the speed of the feed roller **112** when the preceding medium is thick.

(190) The processing in either of steps **S307** and **S311** may be omitted. The control module **151** may change the speed of the feed roller **112** based on both the size and the thickness of a medium. In that case, the control module **151** decreases the speed of the feed roller **112** as the length of a preceding medium decreases and decreases the speed of the feed roller **112** as the thickness of the preceding medium decreases.

(191) As described in detail above, the medium conveying apparatus can control medium feeding better when changing the speed of the feed roller **112** based on the size or the thickness of a medium as well.

(192) FIG. **20** and FIG. **21** are flowcharts illustrating an operation example of medium reading process in a medium conveying apparatus according to yet another embodiment.

(193) The flowcharts illustrated in FIG. **20** and FIG. **21** are executed in place of the flowcharts

illustrated in FIG. 8 and FIG. 9. Processing in steps S401 to S407, S409 to S411, S413 to S417, and S422 to S426 in FIG. 20 and FIG. 21 is similar to the processing in steps S101 to S107, S109 to S111, S114 to S118, and S119 to S123 in FIGS. 8 and 9, and therefore description thereof is omitted. Only steps S408, S412, and S418 to S421 will be described below.

(194) After waiting until the front edge of the medium passes the position of the conveyance roller 119 in step S407, the control module 151 controls the first motor 131 to reduce the speed of the feed roller 112 without stopping the feed roller 112 (step S408). In other words, when the front edge of the medium passes the conveyance roller 119, the control module 151 controls the first motor 131 to cause the feed roller 112 to continue rotation while reducing the rotation speed.

(195) FIG. 22 is a graph for illustrating changes in the speed of the feed roller 112 in a high-speed mode.

(196) A graph G61 in FIG. 22 illustrates an example of changes in the speed of the feed roller 112. In the graph G61, the horizontal axis indicates time, and the vertical axis indicates speed. The speeds of a brake roller 113 and a conveyance roller 119 change similarly as the graphs G12 and G13 illustrated in FIG. 10. Graphs G64 and G65 illustrate changes in the signal values of a fifth medium sensor 118 and a sixth medium sensor 121. In each of the graphs G64 and G65, the horizontal axis indicates time, and the vertical axis indicates the signal value. Times T1 to T6 indicate times similar to the times T1 to T6 indicated in FIG. 10. However, in the examples illustrated in the graphs G61, G64, and G65, the distance between the rear edge of a preceding medium and the front edge of a succeeding medium is shorter, and the time period between the time T4 and the time T5 is shorter, compared with the examples illustrated in the graphs in FIG. 10.

(197) As illustrated in the graph G61 between the time T3 and the time T4, when the front edge of the medium passes the conveyance roller 119, the speed of the feed roller 112 is reduced but is not 0. Consequently, the control module 151 can shorten the distance between successively fed media and reduce the conveyance time of the media while suppressing occurrence of jamming of a medium between the feed roller 112 and the conveyance roller 119.

(198) After waiting until the rear edge of the medium passes the position of the fifth medium sensor 118 in step S410, when determining that a medium remains on a loading tray 103 in step S411, the control module 151 changes the speed of the feed roller 112 (step S412). The control module 151 sets the speed of the feed roller 112 to the first intermediate speed V2a when the conveyance mode is set to the high-speed mode, similarly to step S112 in FIG. 9.

(199) After waiting until the front edge of the medium passes the position of the fifth medium sensor 118 in step S417, the control module 151 calculates the time elapsed after the rear edge of the preceding medium passes a first position before the front edge of the succeeding medium passes a second position (step S418). The time elapsed after the rear edge of the preceding medium passes the first position before the front edge of the succeeding medium passes the second position may be hereinafter referred to as a medium interval. For example, the first position and the second position are set to the position of the fifth medium sensor 118. In that case, the control module 151 calculates the time period from the time when the rear edge of the preceding medium has passed the position of the fifth medium sensor 118 in step S410 to the time when the front edge of the succeeding medium has passed the position of the fifth medium sensor 118 in step S417 as the medium interval. The first position and the second position may be any other position, such as the position of a second medium sensor 114. The first position and the second position are not limited to the same positions and may be positions different from each other.

(200) Next, the control module 151 determines whether the calculated medium interval is less than or equal to a predetermined time (step S418). When the medium interval is greater than the predetermined time, the control module 151 proceeds the processing to step S422.

(201) When the medium interval is less than or equal to the predetermined time, the control module 151 determines a deceleration time in which the feed roller 112 is decelerated, based on the medium interval (step S419). The deceleration time is the time elapsed after the feed roller 112 is

temporarily stopped before rotation of the feed roller **112** is resumed or the time elapsed after the feed roller **112** is decelerated before the feed roller **112** is accelerated.

(202) The control module **151** determines the deceleration time in such a way that the deceleration time increases as the medium interval decreases and the deceleration time decreases as the medium interval increases. The medium conveying apparatus **100** previously stores a table defining a relation between the medium interval and the deceleration time into a storage device **140**, and the control module **151** identifies the deceleration time related to the medium interval with reference to the table stored in the storage device **140**.

(203) Next, the control module **151** reduces the speed of the feed roller **112** (step **S420**).

(204) As illustrated in the graph **G61** in FIG. **22**, when the medium interval (a time period from the time **T4** to the time **T5**) is short, the control module **151** reduces the speed of the feed roller **112** to a predetermined speed at the time **T5** when the front edge of the succeeding medium has passed the position of the fifth medium sensor **118**. For example, the control module **151** changes the speed of the feed roller **112** to a speed at least lower than the second intermediate speed **V2b** and decelerates the feed roller **112** as indicated by a solid line **L2** in the graph **G61**. The control module **151** may change the speed of the feed roller **112** to 0 and stop the feed roller **112** as indicated by a dotted line **D2** in the graph **G61**.

(205) Next, the control module **151** waits until the deceleration time determined in step **S420** elapses (step **S421**).

(206) Next, the control module **151** changes the speed of the feed roller **112** (step **S422**).

(207) As illustrated in FIG. **22**, the control module **151** changes the speed of the feed roller **112** to the final speed **V3a** when the conveyance mode is set to the high-speed mode, similarly to the processing in step **S119** in FIG. **9**.

(208) Thus, the control module **151** temporarily stops or decelerates the feed roller **112** when the medium interval is less than or equal to the predetermined time. Consequently, the control module **151** can suppress occurrence of collision between successively fed media.

(209) The control module **151** temporarily stops or decelerates the feed roller **112** when the front edge of a medium succeeding a preceding medium passes the fifth medium sensor **118**. Then, the control module **151** determines a timing at which rotation of the feed roller **112** is resumed after the feed roller **112** is temporarily stopped or a timing at which the feed roller **112** is accelerated after the feed roller **112** is decelerated, based on the medium interval. Consequently, the control module **151** can suitably set a time period for decelerating the feed roller **112** and can suppress an excessive increase in the medium conveyance time while suppressing occurrence of collision between successively fed media.

(210) The control module **151** may control the first motor **131** to stop the feed roller **112** in step **S408**, similarly to step **S108** in FIG. **8**. In that case, the control module **151** resumes rotation of the feed roller **112** in step **S412** by setting the speed of the feed roller **112** and redriving the first motor **131**, similarly to steps **S112** and **S113** in FIG. **9**.

(211) As described in detail above, the medium conveying apparatus can more control medium feeding better when changing the speed of the feed roller **112** based on intervals between successively fed media as well.

(212) FIG. **23** is a schematic diagram for illustrating driving sources of the feed roller **112**, the brake roller **113**, the conveyance roller **119**, the first facing roller **120**, the ejection roller **123**, and/or the second facing roller **124** in a medium conveying apparatus according to yet another embodiment.

(213) As illustrated in FIG. **23**, the medium conveying apparatus according to the present embodiment includes a first motor **231** in place of the first motor **131**. Further, the medium conveying apparatus includes a second transmission mechanism **132b** in place of the second transmission mechanism **132a** for a second motor **132**.

(214) The first motor **231** is provided in the upper housing **102**, is connected to the feed roller **112**

and the brake roller **113** through a first transmission mechanism **231a**, and drives the brake roller **113** along with the feed roller **112**. The first motor **231** generates a driving force for driving the feed roller **112** and the brake roller **113** in accordance with a control signal from the processing circuit **150**. The first transmission mechanism **231a** includes one or a plurality of pulleys, belts, gears, etc., that are provided between the first motor **231**, and the shaft **113a** of the brake roller **113** and the shaft **112a** of the feed roller **112**. The first transmission mechanism **231a** transmits the driving force generated by the first motor **231** to the feed roller **112** and the brake roller **113**. In particular, one or a plurality of gears for changing the rotation direction and the rotation speed of each roller are provided between the shaft **113a** of the brake roller **113** and the shaft **112a** of the feed roller **112**. The gears are located outside the medium conveyance path in the width direction **A8** to transmit the driving force with the medium conveyance path in between. Consequently, the first motor **231** feeds a medium by rotating the feed roller **112** and the brake roller **113**. The first motor **231** is an example of a driving source of the brake roller **113**.

(215) The first motor **231** can reliably transmit the driving force to the brake roller **113** before the feed roller **112** since the first motor **231** is provided in the housing on the brake roller **113** side (the upper housing **102**). Consequently, the first motor **231** can start rotation of the brake roller **113** before starting rotation of the feed roller **112** and can stably separate a medium. Further, it is preferable that the first motor **231** is directly connected to the shaft **113a** of the brake roller **113** by a belt. Consequently, the first motor **231** can more reliably start rotation of the brake roller **113** before starting rotation of the feed roller **112**.

(216) A second motor **132** is provided in the upper housing **102** separately from the first motor **231**, is connected to the conveyance roller **119** and the ejection roller **123** through the second transmission mechanism **132b**, and drives the conveyance roller **119** and the ejection roller **123**. The second motor **132** generates a driving force for driving the conveyance roller **119** and the ejection roller **123** in accordance with a control signal from the processing circuit **150**. The second transmission mechanism **132b** includes one or a plurality of pulleys, belts, gears, etc., that are provided between the second motor **132**, a shaft **119a** of the conveyance roller **119** and a shaft **123a** of the ejection roller **123**. The second transmission mechanism **132b** transmits the driving force generated by the second motor **132** to the conveyance roller **119** and the ejection roller **123**. Consequently, the second motor **132** causes the conveyance roller **119** and the ejection roller **123** to convey and eject a medium by rotating the conveyance roller **119** and the ejection roller **123**.

(217) FIG. **24** and FIG. **25** are flowcharts illustrating an operation example of medium reading process in a medium conveying apparatus according to the yet another embodiment.

(218) The flowcharts illustrated in FIG. **24** and FIG. **25** are executed in place of the flowcharts illustrated in FIG. **8** and FIG. **9**. Processing in steps **S501** to **S507**, **S509** to **S511**, **S514**, **S516** to **S518**, and **S520** to **S523** in FIG. **24** and FIG. **25** is similar to the processing in steps **S101** to **S107**, **S109** to **S111**, **S114**, **S116** to **S118**, and **S120** to **S123** in FIGS. **8** and **9**, and therefore description thereof is omitted. Only steps **S508**, **S512**, **S513**, **S515**, and **S519** will be described below.

(219) After waiting until the front edge of the medium passes the position of the conveyance roller **119** in step **S507**, the control module **151** controls the first motor **231** to stop or decelerate the brake roller **113** along with the feed roller **112** (step **S508**). The medium conveying apparatus **100** has a first mode giving priority to securing the distance between media and a second mode giving priority to preventing occurrence of multi feed of media. The first mode and the second mode are set by a user by using the operation device **105** or the information processing apparatus.

(220) FIG. **26** is a graph for illustrating changes in the speeds of a feed roller, a brake roller, and a conveyance roller in a high-speed mode.

(221) In FIG. **26**, a graph **G71** illustrates an example of changes in the speed of the feed roller **112**, a graph **G72** illustrates an example of changes in the speed of the brake roller **113**, and a graph **G73** illustrates an example of changes in the speed of the conveyance roller **119**. In each of the graphs **G71** to **G73**, the horizontal axis indicates time, and the vertical axis indicates speed. Graphs **G14**

and G15 illustrate changes in the signal values of a fifth medium sensor **118** and a sixth medium sensor **121**, and change similarly to the graphs G14 and G15 illustrated in FIG. 10. Times T1 to T6 indicate times similar to the times T1 to T6 indicated in FIG. 10.

(222) When set to the first mode, the control module **151** controls the first motor **231** to stop the feed roller **112** and the brake roller **113**. In this case, as indicated by a solid line L3 in the graph G71 and a solid line L4 in the graph G72, the speeds of the feed roller **112** and the brake roller **113** become 0 between the time T3 and the time T4. Consequently, the control module **151** can reliably suppress occurrence of jamming of a medium between the feed roller **112** and the conveyance roller **119**.

(223) When set to the second mode, the control module **151** controls the first motor **231** to cause the feed roller **112** and the brake roller **113** to continue rotation while reducing the rotation speeds (without stopping). In this case, as indicated by a dotted line D3 in the graph G71 and a dotted line D4 in the graph G72, the speeds of the feed roller **112** and the brake roller **113** do not become 0 but are reduced between the time T3 and the time T4. Consequently, the control module **151** can suppress reduction in separation performance due to stopping the brake roller **113** and can suppress occurrence of multi feed of media.

(224) When determining that a medium remains on the loading tray **103** in step S511 after waiting until the rear edge of the medium passes the position of the fifth medium sensor **118** in step S510, the control module **151** sets the speed of the feed roller **112** for feeding a succeeding medium (step S512).

(225) When being set to the first mode, the control module **151** sets the speed of the feed roller **112** to the first intermediate speed V2a at the time T4 as illustrated in the graph G71 in FIG. 26.

Further, the control module **151** sets the speed of the brake roller **113** to a first intermediate speed U2a at the time T4 as illustrated in the graph G72. The first intermediate speed U2a is set to a speed higher than the initial speed U1 and lower than the final speed U3a.

(226) Next, the control module **151** controls the first motor **231** to rotate the feed roller **112** and the brake roller **113** at the set speeds (step S513). When being set to the first mode and stopping the feed roller **112** and the brake roller **113**, the control module **151** resumes rotation of the feed roller **112** and the brake roller **113** to feed and convey a succeeding medium.

(227) When set to the second mode and not stopping the feed roller **112** and the brake roller **113**, the control module **151** changes the speeds of the feed roller **112** and the brake roller **113**. The control module **151** increases the speed of the feed roller **112** to the first intermediate speed V2a and increases the speed of the brake roller **113** to the first intermediate speed U2a.

(228) After waiting until the front edge of the succeeding medium passes the position of the second medium sensor **114** in step S514, the control module **151** changes the speeds of the feed roller **112** and the brake roller **113** (step S515).

(229) The control module **151** changes the speed of the feed roller **112** to the second intermediate speed V2b as illustrated in the graph G71 in FIG. 26. Further, the control module **151** changes the speed of the brake roller **113** to a second intermediate speed U2b as illustrated in the graph G72. The second intermediate speed U2b is set to a speed higher than the first intermediate speed U2a and lower than the final speed U3a.

(230) After waiting until the front edge of the succeeding medium passes the position of the fifth medium sensor **118** in step S518, the control module **151** changes the speeds of the feed roller **112** and the brake roller **113** (step S519).

(231) The control module **151** changes the speed of the feed roller **112** to a final speed V3a at the time T5 as illustrated in the graph G71 in FIG. 26. Further, the control module **151** changes the speed of the brake roller **113** to the final speed U3a as illustrated in the graph G72.

(232) In step S508, the control module **151** may stop the feed roller **112** by controlling the first electromagnetic clutch **133** to interrupt the driving force from the first motor **231** to the feed roller **112**, while decelerating the brake roller **113**. In other words, when the front edge of the medium

passes the conveyance roller **119**, the control module **151** continues to transmit the driving force from the first motor **231** to the brake roller **113** while interrupting the driving force from the first motor **231** to the feed roller **112** by controlling the first electromagnetic clutch **133**. In that case, the control module **151** resumes rotation of the feed roller **112** in step **S513** by controlling the first electromagnetic clutch **133** to transmit the driving force from the first motor **231** to the feed roller **112**. Consequently, the control module **151** can suppress decrease in the distance between successively fed media while suppressing degradation in separation performance due to stopping the brake roller **113**. The medium conveying apparatus may interrupt the driving force from the first motor **231** to the feed roller **112** by a mechanism, such as a planetary gear mechanism, other than the electromagnetic clutch. The planetary gear is provided to be able to move between a position at which the planetary gear transmits the driving force from the first motor **231** to the feed roller **112** and a position at which the planetary gear does not transmit the driving force from the first motor **231** to the feed roller **112**.

(233) According to the present embodiment, the feed roller **112** and the brake roller **113** are driven by the same first motor **231**, and therefore the speed of the feed roller **112** and the speed of the brake roller **113** have a proportional relation. The speed of the brake roller **113** is set to be lower than the speed of the feed roller **112** and higher than $\frac{1}{2}$ of the speed of the feed roller **112**.

(234) The medium conveying apparatus may operate in either mode of the first mode and the second mode in a fixed manner.

(235) As described in detail above, the medium conveying apparatus can control medium feeding better when the feed roller **112** and the brake roller **113** are driven by the same first motor **231** as well.

(236) FIG. **27** is a schematic diagram for illustrating driving sources of a feed roller, a brake roller, a conveyance roller, a first facing roller, an ejection roller, and/or a second facing roller in a medium conveying apparatus according to yet another embodiment.

(237) As illustrated in FIG. **27**, the medium conveying apparatus according to the present embodiment includes the conveyance roller **219**, the first facing roller **220**, the ejection roller **223**, and the second facing roller **224** in place of the conveyance roller **119**, the first facing roller **120**, the ejection roller **123**, and the second facing roller **124**. The structures of the conveyance roller **219**, the first facing roller **220**, the ejection roller **223**, and the second facing roller **224** are similar to the structures of the conveyance roller **219**, the first facing roller **220**, the ejection roller **223**, and the second facing roller **224** in the medium conveying apparatus illustrated in FIG. **15**. Further, the medium conveying apparatus includes the second motor **232** in place of the second motor **132**. The structure of the second motor **232** is similar to the structure of the second motor **232** in the medium conveying apparatus illustrated in FIG. **15**. Further, the medium conveying apparatus includes the first motor **231** in place of the first motor **131**. The structure of the first motor **231** is similar to the structure of the first motor **231** in the medium conveying apparatus illustrated in FIG. **23**.

(238) In other words, the first motor **231** drives the brake roller **113** along with the feed roller **112**, and the second motor **232** drives the conveyance roller **219** and the ejection roller **223** in the medium conveying apparatus according to the present embodiment, similarly to the medium conveying apparatus illustrated in FIG. **23**.

(239) As described in detail above, the medium conveying apparatus can control medium feeding better when the feed roller **112** and the brake roller **113** are driven by the same first motor **231** while providing the conveyance roller **219** in the lower housing **101** as well.

(240) FIG. **28** is a schematic diagram for illustrating driving sources of a feed roller, a brake roller, a conveyance roller, a first facing roller, an ejection roller, and/or a second facing roller in a medium conveying apparatus according to yet another embodiment.

(241) As illustrated in FIG. **28**, the medium conveying apparatus according to the present embodiment includes a third motor **336** in addition to the first motor **131**. The medium conveying apparatus includes a first transmission mechanism **131b** in place of the first transmission

mechanism **131a** for the first motor **131** and includes a second transmission mechanism **132b** in place of the second transmission mechanism **132a** for the second motor **132**. The structure of the second motor **132** is similar to the structure of the second motor **132** illustrated in FIG. **23**.

(242) The first motor **131** is connected to the feed roller **112** through the first transmission mechanism **131b** and drives the feed roller **112**. The first motor **131** generates a driving force for driving the feed roller **112** in accordance with a control signal from the processing circuit **150**. The first transmission mechanism **131b** includes one or a plurality of pulleys, belts, gears, etc., that are provided between the first motor **131** and a shaft **112a** of the feed roller **112** and transmits the driving force generated by the first motor **131** to the feed roller **112**. Consequently, the first motor **131** feeds a medium by rotating the feed roller **112**.

(243) The third motor **336** is provided in the upper housing **102** separately from the first motor **131** and the second motor **132**, is connected to the brake roller **113** through a third transmission mechanism **336a**, and drives the brake roller **113**. The third motor **336** generates a driving force for driving the brake roller **113** in accordance with a control signal from the processing circuit **150**. The third transmission mechanism **336a** includes one or a plurality of pulleys, belts, gears, etc., that are provided between the third motor **336** and a shaft **113a** of the brake roller **113** and transmits the driving force generated by the third motor **336** to the brake roller **113**. Consequently, the third motor **336** feeds a medium by rotating the brake roller **113**. The third motor **336** is an example of a driving source of the brake roller **113**.

(244) FIG. **29** is a flowchart illustrating an example of part of operation of medium reading process in a medium conveying apparatus according to the yet another embodiment.

(245) The flowchart illustrated in FIG. **29** is executed in place of the flowchart illustrated in FIG. **9**. Processing in steps **S611**, **S614**, **S615**, **S616**, **S618**, **S620**, and **S622** to **S625** in FIG. **29** is similar to the processing in steps **S111**, **S112**, **S113**, **S116**, **S117**, **S118**, and **S120** to **S123** in FIG. **9**, and therefore description thereof is omitted. Only steps **S612**, **S613**, **S617**, **S619**, and **S621** will be described below.

(246) After waiting until the rear edge of the medium passes the position of the fifth medium sensor **118** in step **S110**, when determining that a medium remains on the loading tray **103** in step **S611**, the control module **151** controls the second motor **132** to reduce the speed of the conveyance roller **119** (step **S612**). In other words, the control module **151** reduces the rotation speed of the conveyance roller **119** when the rear edge of the medium passes the fifth medium sensor **118**.

(247) FIG. **30** is a graph for illustrating changes in the speeds of the feed roller **112**, the brake roller **113**, and the conveyance roller **119** in the high-speed mode.

(248) In FIG. **30**, a graph **G81** illustrates an example of changes in the speed of the feed roller **11**, a graph **G82** illustrates an example of changes in the speed of the brake roller **113**, and a graph **G83** illustrates an example of changes in the speed of the conveyance roller **119**. In each of the graphs **G81** to **G83**, the horizontal axis indicates time, and the vertical axis indicates speed. Graphs **G14** and **G15** illustrate changes in the signal values of the fifth medium sensor **118** and the sixth medium sensor **121** and change similarly to the graphs **G14** and **G15** illustrated in FIG. **10**. Times **T1** to **T6** indicate times similar to the times **T1** to **T6** indicated in FIG. **10**. The control module **151** may reduce the rotation speed of the conveyance roller **119** when the rear edge of the medium passes the second medium sensor **114** instead of the fifth medium sensor **118**.

(249) As illustrated in the graph **G83**, the control module **151** reduces the speed of the conveyance roller **119** when the rear edge of the medium passes the fifth medium sensor **118** at the time **T4**.

During separation, a force attempting to keep a medium on the upstream side is exerted on the medium by the feed roller **112** and the brake roller **113**. The force disappears when the rear edge of the medium leaves the feed roller **112** and the brake roller **113**, and therefore the conveyance speed of the medium tends to increase. By reducing the speed of the conveyance roller **119** when the rear edge of the medium passes the feed roller **112** and the brake roller **113**, the control module **151** can convey the medium at a stable speed.

(250) Next, the control module **151** waits until a sixth predetermined time elapses (step **S613**). The sixth predetermined time is set to a time by which medium feeding by the feed roller **112** can be delayed for the amount of delay in medium conveyance due to deceleration of the conveyance roller **119**. Consequently, the control module **151** can delay medium feeding by the feed roller **112** for the amount of delay in medium conveyance by the conveyance roller **119** and can convey the medium at a stable speed. The processing in step **S613** may be omitted.

(251) After waiting until the rear edge of the medium passes the imaging position in step **S616**, the control module **151** controls the second motor **132** to increase the speed of the conveyance roller **119** (step **S617**). In other words, the control module **151** increases the rotation speed of the conveyance roller **119** when the rear edge of the medium passes the imaging position of the imaging device **122**.

(252) As illustrated in the graph **G83** in FIG. **30**, the control module **151** increases the speed of the conveyance roller **119** when the rear edge of the medium passes the imaging position between the time **T4** and the time **T5**. By changing the speed of the conveyance roller **119** after the rear edge of the medium passes the imaging position, the control module **151** can suppress occurrence of expansion or contraction within an input image. Further, when the rear edge of the medium passes the conveyance roller **119** and a first facing roller **120**, the conveyance force exerted on the medium decreases, and the conveyance speed of the medium tends to decrease. By increasing the speed of the ejection roller **123** after the rear edge of the medium passes the conveyance roller **119** and the first facing roller **120**, the control module **151** can convey the medium at a stable speed.

(253) After acquiring an input image in step **S618**, the control module **151** controls the first motor **131** to increase the speed of the feed roller **112** (step **S619**). In other words, the control module **151** increases the rotation speed of the feed roller **112** when the rear edge of the medium passes the conveyance roller **119**.

(254) As indicated by a solid line **L5** in the graph **G81** in FIG. **30**, the control module **151** increases the speed of the feed roller **112** to the second intermediate speed **V2b** when the rear edge of the medium passes the position of the conveyance roller **119** (after the signal value in the graph **G15** changes from **H** to **L**). In particular, the control module **151** increases the speed of the feed roller **112** after increasing the speed of the conveyance roller **119**. Consequently, the control module **151** can suppress occurrence of jamming of the medium due to the medium being pushed by the feed roller **112** and being bent between the feed roller **112** and the conveyance roller **119**.

(255) As indicated by a dotted line **D5** in the graph **G81**, the control module **151** may increase the speed of the feed roller **112** to the final speed **V3a** higher than the second intermediate speed **V2b** after the rear edge of the medium passes the position of the conveyance roller **119**. Consequently, the control module **151** can make up for the delay in medium conveyance by the conveyance roller **119** and the feed roller **112**.

(256) After waiting until the front edge of the medium passes the fifth medium sensor **118** in step **S620**, the control module **151** controls the first motor **131** to increase the speed of the feed roller **112** (step **S621**).

(257) As indicated by a solid line **L6** in the graph **G81** in FIG. **30**, the control module **151** increases the speed of the feed roller **112** to the final speed **V3a** at the time **T5**. As indicated by a dotted line **D6** in the graph **G81**, the control module **151** may increase the speed of the feed roller **112** to the final speed **V3d** higher than the final speed **V3a** at the time **T5**. Consequently, the control module **151** can make up for the delay in medium conveyance by the conveyance roller **119** and the feed roller **112**. When changing the speed of the feed roller **112** to the final speed **V3d**, the control module **151** may change the speed of the feed roller **112** to the final speed **V3a** before the front edge of the succeeding medium passes the position of the fifth medium sensor **118**, as indicated by a dotted line **D7** in the graph **G81**. Consequently, the control module **151** can suppress occurrence of jamming of the medium due to the medium being pushed by the feed roller **112** and being bent between the feed roller **112** and the conveyance roller **119**.

(258) According to the present embodiment, the driving source of the feed roller **112**, the driving source of the brake roller **113**, and the driving source of the conveyance roller **119** and the ejection roller **123** are separately provided, and therefore the speeds of the rollers can be changed at timings independent of each other. Therefore, the control module **151** can flexibly control feeding and conveyance of a medium.

(259) As indicated by a dotted line D8 in the graph G82, the control module **151** may temporarily stop the brake roller **113** while the rear edge of the medium is passing the nip region of the feed roller **112** and the brake roller **113** and resume rotation of the brake roller **113** after the rear edge passes the nip region. Alternatively, as indicated by a dotted line D9 in the graph G82, the control module **151** may reduce the speed of the brake roller **113** while the rear edge of the medium is passing the nip region, increase the speed of the brake roller **113** after the rear edge passes the nip region, and return to the final speed U3a. When the rear edge of the preceding medium passes the nip region of the feed roller **112** and the brake roller **113**, the brake roller **113** may rotate in a direction opposite to the medium feeding direction with force due to reduction in a load exerted on the brake roller **113** in a medium feeding direction by the medium. At that time, a succeeding medium may be pushed back to the upstream side due to elastically deformed parts of the feed roller **112** and the brake roller **113** returning to the original shapes. In that case, jamming of a medium may occur due to the front edge of the succeeding medium being pushed up (turned up) by the brake roller **113**. By reducing the speed of the brake roller **113** when the rear edge of a medium passes the nip region, the control module **151** can suppress occurrence of jamming of the succeeding medium due to the front edge of the medium being turned up.

(260) The control module **151** may start rotation of the brake roller **113** before rotation of the feed roller **112**. Consequently, the control module **151** can suppress avalanching a plurality of media placed on the loading tray **103** into between the feed roller **112** and the brake roller **113** and can suppress the occurrence of multi feed of media.

(261) As described in detail above, the medium conveying apparatus can control medium feeding better when the speed of the conveyance roller **119** is changed while the feed roller **112**, the brake roller **113**, and the conveyance roller **119** are driven by separate motors as well.

(262) The medium conveying apparatus may drive the brake roller **113** by either the first motor or the second motor when changing the speed of the conveyance roller **119**. In that case, the control module **151** adjusts the timing of changing the speed of the brake roller **113** to the timing of changing the speed of the feed roller **112** or the conveyance roller **119**. Further, the control module **151** may change the speed of the feed roller **112** or the brake roller **113** by controlling the first electromagnetic clutch **133** or the second electromagnetic clutch **134**. Further, the medium conveying apparatus may include a driving force interruption mechanism, such as an electromagnetic clutch, between the conveyance roller **119** and the second motor, and the control module **151** may change the speed of the conveyance roller **119** by controlling the driving force interruption mechanism.

(263) FIG. **31** is a schematic diagram for illustrating driving sources of a feed roller, a brake roller, a conveyance roller, a first facing roller, an ejection roller, and/or a second facing roller in a medium conveying apparatus according to yet another embodiment.

(264) As illustrated in FIG. **31**, the medium conveying apparatus according to the present embodiment includes the conveyance roller **219**, the first facing roller **220**, the ejection roller **223**, and the second facing roller **224** in place of the conveyance roller **119**, the first facing roller **120**, the ejection roller **123**, and the second facing roller **124**. The structures of the conveyance roller **219**, the first facing roller **220**, the ejection roller **223**, and the second facing roller **224** are similar to the structures of the conveyance roller **119**, the first facing roller **120**, the ejection roller **123**, and the second facing roller **124** in the medium conveying apparatus illustrated in FIG. **15**. Further, the medium conveying apparatus includes the second motor **232** in place of the second motor **132**. The structure of the second motor **232** is similar to the structure of the second motor **132** in the medium

conveying apparatus illustrated in FIG. 15. Further, the medium conveying apparatus includes the third motor **336** in addition to the first motor **131**. The structures of the first motor **131** and the third motor **336** are similar to the structures of the first motor **131** and the third motor **336** in the medium conveying apparatus illustrated in FIG. 28.

(265) In other words, the first motor **131** drives the feed roller **112**, the third motor **336** drives the brake roller **113**, and the second motor **232** drives the conveyance roller **219** and the ejection roller **223** in the medium conveying apparatus according to the present embodiment, similarly to the medium conveying apparatus illustrated in FIG. 28.

(266) As described in detail above, the medium conveying apparatus can control medium feeding better when the feed roller **112**, the brake roller **113**, and the conveyance roller **219** are driven by separate motors while providing the conveyance roller **219** in the lower housing **101**.

(267) FIG. 32 is a flowchart illustrating an example of part of operation of medium reading process in a medium conveying apparatus according to yet another embodiment. The medium conveying apparatus according to the present embodiment includes the driving sources illustrated in FIG. 28 or the driving sources illustrated in FIG. 31.

(268) The flowchart illustrated in FIG. 32 is executed in place of the flowchart illustrated in FIG. 9. Processing in steps **S711**, **S714** to **S716**, **S719** to **S721**, **S724**, and **S727** to **S731** in FIG. 32 is similar to the processing in steps **S111**, **S112** to **S114**, **S115** to **S117**, **S118**, and **S119** to **S123** in FIG. 9, and therefore description thereof is omitted. Only steps **S712**, **S713**, **S717**, **S718**, **S722**, **S723**, **S725**, and **S726** will be described below. When the flowchart illustrated in FIG. 32 is executed, steps **S301** to **S312** in the flowchart illustrated in FIG. 18 are executed in place of steps **S101** to **S110** in the flowchart illustrated in FIG. 8.

(269) After waiting until the rear edge of the medium passes the position of the fifth medium sensor **118** in step **S311**, the control module **151** determines whether the size of the preceding medium is greater than the first size threshold value and whether the size is greater than the second size threshold value (step **S712**). When the size of the preceding medium is less than or equal to the first size threshold value and greater than or equal to the second size threshold value, the control module **151** proceeds the processing to step **S714** without particularly executing processing.

(270) When the size of the preceding medium is greater than the first size threshold value or less than the second size threshold value, the control module **151** waits until a seventh predetermined time elapses (step **S713**). For example, the seventh predetermined time is set to a time acquired by subtracting the time required for medium conveyance from the time required for image processing when a medium greater than the first size threshold value is conveyed. Alternatively, the seventh predetermined time may be set to the maximum value of the stopping time of the feed roller **112** when collision of a medium occurs in a previously performed experiment of successively conveying media smaller than the second size threshold value while changing the stopping time. Alternatively, the seventh predetermined time may be set to a time by which medium feeding by the feed roller **112** can be delayed for the amount of delay in medium ejection due to deceleration of an ejection roller **123** in the processing described later.

(271) In other words, the control module **151** stops the feed roller **112** in step **S309** when the front edge of the medium passes a conveyance roller **119** in step **S308**. Further, the control module **151** resumes feeding by the feed roller **112** in step **S715** after the rear edge of the medium passes the position of the fifth medium sensor **118** in step **S311**. The control module **151** delays the resumption timing of feeding by the feed roller **112** when the size of the medium of which the rear edge has passed the position of the fifth medium sensor **118** is greater than the first size threshold value relative to the resumption timing of feeding by the feed roller **112** when the size of the medium of which the rear edge has passed the position of the fifth medium sensor **118** is less than or equal to the first size threshold value. Similarly, the control module **151** delays the resumption timing of feeding by the feed roller **112** when the size of the medium of which the rear edge has passed the position of the fifth medium sensor **118** is less than the second size threshold value

relative to the resumption timing of feeding by the feed roller **112** when the size of the medium of which the rear edge has passed the position of the fifth medium sensor **118** is greater than or equal to the second size threshold value.

(272) FIG. **33** is a graph for illustrating changes in the speeds of the feed roller **112**, the brake roller **113**, and the conveyance roller **119** in a high-speed mode when the size of a preceding medium is greater than the first size threshold value or less than the second size threshold value.

(273) In FIG. **33**, a graph G91 illustrates an example of changes in the speed of the feed roller **112**, a graph G92 illustrates an example of changes in the speed of the brake roller **113**, and a graph G93 illustrates an example of changes in the speed of the conveyance roller **119**. In each of the graphs G91 to G93, the horizontal axis indicates time, and the vertical axis indicates speed. Graphs G14 and G15 illustrate changes in the signal values of the fifth medium sensor **118** and a sixth medium sensor **121** and change similarly to the graphs G14 and G15 illustrated in FIG. **10**. Times T1 to T6 indicate times similar to the times T1 to T6 indicated in FIG. **10**.

(274) As illustrated in the graph G91, when the size of a preceding medium is greater than the first size threshold value or less than the second size threshold value, the control module **151** delays the resumption timing of feeding by the feed roller **112** at the time T4. When a large-sized medium is conveyed, the size of an input image increases, and a considerable time may be required for image processing. By delaying the resumption timing of feeding when a large-sized medium is conveyed, the control module **151** can smoothly convey the medium and acquire a satisfactory image without stopping the medium for image processing. Further, as described above, the distance between successively conveyed media generally tends to decrease as the length of the media decreases. By delaying the resumption timing of feeding when a small-sized medium is conveyed, the control module **151** can suppress occurrence of collision between successively conveyed media. Further, when a large-sized medium or a small-sized medium is conveyed, the control module **151** reduces the speed of the ejection roller **123** in the processing described later. In that case, the control module **151** can convey the medium at a stable speed by delaying the resumption timing of feeding.

(275) After waiting until the front edge of the medium passes the position of the second medium sensor **114** in step S716, the control module **151** determines whether the size of the preceding medium is greater than the first size threshold value and whether the size is less than the second size threshold value (step S717). When the size of the preceding medium is less than or equal to the first size threshold value and greater than or equal to the second size threshold value, the control module **151** proceeds the processing to step S719 without executing particular processing.

(276) When the size of the preceding medium is greater than the first size threshold value or less than the second size threshold value, the control module **151** waits until the seventh predetermined time elapses (step S718).

(277) As illustrated in the graph G91 in FIG. **33**, when the size of the preceding medium is greater than the first size threshold value or less than the second size threshold value, the control module **151** delays the timing of increasing the speed of the feed roller **112** between the time T4 and the time T5. Consequently, the control module **151** can smoothly convey a large-sized medium and acquire a satisfactory image without stopping the medium for image processing. Further, when a small-sized medium is conveyed, the control module **151** can suppress occurrence of collision with another medium. Further, the control module **151** can convey a medium at a stable speed.

(278) After acquiring an input image step S721, the control module **151** determines whether the size of the preceding medium is greater than the first size threshold value and whether the size is less than the second size threshold value (step S722). When the size of the preceding medium is less than or equal to the first size threshold value and greater than or equal to the second size threshold value, the control module **151** proceeds the processing to step S724 without particularly executing processing.

(279) When the size of the preceding medium is greater than the first size threshold value or less than the second size threshold value, the control module **151** controls the second motor **132** to

reduce the speeds of the conveyance roller **119** and the ejection roller **123** (step **S723**).

(280) In other words, the control module **151** sets the rotation speed of the ejection roller **123** when the size of the medium is greater than the first size threshold value to a speed lower than the rotation speed of the ejection roller **123** when the size of the medium is less than or equal to the first size threshold value. Further, the control module **151** sets the rotation speed of the ejection roller **123** when the size of the medium is less than the second size threshold value to a speed lower than the rotation speed of the ejection roller **123** when the size of the medium is greater than or equal to the second size threshold value.

(281) As illustrated in the graph **G93** in **FIG. 33**, when the size of the preceding medium is greater than the first size threshold value or less than the second size threshold value, the control module **151** reduces the speed of the ejection roller **123** between the time **T4** and the time **T5**. When a large-sized medium is ejected, the front edge of the medium may go past the front edge of the ejection tray **104**, and the medium may fly out of the ejection tray **104** by the weight of the front edge going past the ejection tray **104**. By reducing the speed of the ejection roller **123** when a large-sized medium is ejected, the control module **151** can suppress occurrence of flying out of the medium from the ejection tray **104**. Further, when a plurality of media with different sizes are ejected, only a small-sized medium may be ejected with force, and the rear edges of the media may not be aligned. By reducing the speed of the ejection roller **123** when a small-sized medium is ejected, the control module **151** can improve media alignment.

(282) After waiting until the front edge of the medium passes the position of the fifth medium sensor **118** in step **S724**, the control module **151** determines whether the size of the preceding medium is greater than the first size threshold value and whether the size is less than the second size threshold value (step **S725**). When the size of the preceding medium is less than or equal to the first size threshold value and greater than or equal to the second size threshold value, the control module **151** proceeds the processing to step **S727** without executing particular processing.

(283) When the size of the preceding medium is greater than the first size threshold value or less than the second size threshold value, the control module **151** waits until the seventh predetermined time elapses (step **S727**).

(284) In other words, the control module **151** increases the rotation speed of the feed roller **112** in step **S727** after the fifth medium sensor **118** detects the front edge of the medium in step **S724**. The control module **151** delays the timing of increasing the rotation speed of the feed roller **112** when the size of the preceding medium is greater than the first size threshold value relative to the timing of increasing the rotation speed of the feed roller **112** when the size of the preceding medium is less than or equal to the first size threshold value. Similarly, the control module **151** delays the timing of increasing the rotation speed of the feed roller **112** when the size of the preceding medium is less than the second size threshold value relative to the timing of increasing the rotation speed of the feed roller **112** when the size of the preceding medium is greater than or equal to the second size threshold value.

(285) As illustrated in the graph **G91** in **FIG. 33**, when the size of the preceding medium is greater than the first size threshold value or less than the second size threshold value, the control module **151** delays the timing of increasing the speed of the feed roller **112** at the time **T5**. Consequently, the control module **151** can smoothly convey a large-sized medium and acquire a satisfactory image without stopping the medium for image processing. Further, when a small-sized medium is conveyed, the control module **151** can suppress occurrence of collision with another medium. Further, the control module **151** can convey a medium at a stable speed.

(286) One of processing in steps **S712** and **S713**, **S717** and **S718**, **S722** and **S723**, and **S725** and **S726** may be omitted. Further, the control module **151** may determine only either of whether the size of the medium is greater than the first size threshold value and whether the size of the medium is less than the second size threshold value in step **S712**, **S717**, **S722**, or **S725**. In that case, the control module **151** executes the processing in step **S713**, **S718**, **S723**, or **S726** regardless of

whether the size of the medium is less than the second size threshold value or whether the size of the medium is greater than the first size threshold value.

(287) In step **S712**, **S717**, or **S725**, the control module **151** may determine whether the size of a medium preceding the medium of which the rear edge has passed the position of the fifth medium sensor **118** is greater than the first size threshold value and whether the size is less than the second size threshold value. In other words, the control module **151** delays the resumption timing of feeding by the feed roller **112** when the size of a medium preceding the medium of which the rear edge has passed the position of the fifth medium sensor **118** is greater than the first size threshold value relative to the resumption timing of feeding by the feed roller **112** when the size of a medium preceding the medium of which the rear edge has passed the position of the fifth medium sensor **118** is less than or equal to the first size threshold value. Similarly, the control module **151** delays the resumption timing of feeding by the feed roller **112** when the size of a medium preceding the medium of which the rear edge has passed the position of the fifth medium sensor **118** is less than the second size threshold value relative to the resumption timing of feeding by the feed roller **112** when the size of a medium preceding the medium of which the rear edge has passed the position of the fifth medium sensor **118** is greater than or equal to the second size threshold value.

(288) In these cases, the control module **151** can smoothly convey a large-sized medium and acquire a satisfactory image without stopping the medium for image processing as well. Further, when a small-sized medium is conveyed, the control module **151** can suppress occurrence of collision with another medium.

(289) As indicated by a dotted line **D8** in **G92** in **FIG. 33**, the control module **151** may temporarily stop the brake roller **113** while the rear edge of the medium is passing the nip region of the feed roller **112** and the brake roller **113** and resume rotation of the brake roller **113** after the rear edge passes the nip region. Alternatively, as indicated by a dotted line **D9** in **G92**, the control module **151** may reduce the speed of the brake roller **113** while the rear edge of the medium is passing the nip region and increase and return the speed of the brake roller **113** to the final speed **U3a** after the rear edge passes the nip region. Consequently, the control module **151** can suppress occurrence of jamming of a medium due to the front edge of the succeeding medium being turned up.

(290) The control module **151** may start rotation of the brake roller **113** before rotation of the feed roller **112**. Consequently, the control module **151** can suppress a surge of a plurality of media placed on the loading tray **103** into between the feed roller **112** and the brake roller **113** and can suppress occurrence of multi feed of media.

(291) As described in detail above, the medium conveying apparatus can control medium feeding better when the speed of the ejection roller **123** is changed based on the size of the medium as well.

(292) The medium conveying apparatus may drive the brake roller **113** by either a first motor or a second motor when changing the speed of the ejection roller **123**. In that case, the control module **151** adjusts the timing of changing the speed of the brake roller **113** to the timing of changing the speed of the feed roller **112** or the ejection roller **123**. Further, the control module **151** may change the speed of the feed roller **112** or the brake roller **113** by controlling the first electromagnetic clutch **133** or the second electromagnetic clutch **134**. Further, the medium conveying apparatus may include a driving force interruption mechanism, such as an electromagnetic clutch, between the ejection roller **123** and the second motor, and the control module **151** may change the speed of the conveyance roller **119** by controlling the driving force interruption mechanism. The driving force interruption mechanism may be a planetary gear mechanism. The planetary gear is provided to be able to move between a position at which the planetary gear transmits the driving force from the second motor to the ejection roller **123** and a position at which the planetary gear does not transmit the driving force from the second motor to the ejection roller **123**.

(293) **FIG. 34** and **FIG. 35** are flowcharts illustrating an operation example of medium reading process in a medium conveying apparatus according to yet another embodiment. The medium conveying apparatus according to the present embodiment includes the driving sources illustrated

in FIG. 28 or the driving sources illustrated in FIG. 31.

(294) The flowcharts illustrated in FIG. 34 and FIG. 35 are executed in place of the flowcharts illustrated in FIG. 8 and FIG. 9. Processing in steps S801 to S805, S810 to S821, and S823 to S826 in FIG. 34 and FIG. 35 is similar to the processing in steps S101 to S105, S107 to S118, and S120 to S123 in FIG. 8 and FIG. 9, and therefore description thereof is omitted. Only steps S806 to S809 and S822 will be described below.

(295) After waiting until the front edge of the medium passes the position of a fifth medium sensor 118 in step S805, the control module 151 controls a first motor 131 to temporarily stop the feed roller 112 or reduce the speed of the feed roller 112 (step S806).

(296) FIG. 36 is a graph for illustrating changes in the speeds of the feed roller 112, a brake roller 113, and a conveyance roller 119 in a high-speed mode.

(297) In FIG. 36, a graph G101 illustrates an example of changes in the speed of the feed roller 112, a graph G102 illustrates an example of changes in the speed of the brake roller 113, and a graph G103 illustrates an example of changes in the speed of the conveyance roller 119. In each of the graphs G101 to G103, the horizontal axis indicates time, and the vertical axis indicates speed. Graphs G14 and G15 illustrate changes in the signal values of the fifth medium sensor 118 and a sixth medium sensor 121 and change similarly to the graphs G14 and G15 illustrated in FIG. 10. Times T1 to T6 indicate times similar to the times T1 to T6 indicated in FIG. 10.

(298) As indicated by a solid line L10 in the graph G101, the control module 151 stops the feed roller 112 at the time T2. Alternatively, as indicated by a dotted line D10 in the graph G101, the control module 151 decelerates the feed roller 112 at the time T2.

(299) Next, the control module 151 detects the thickness of the conveyed medium, similarly to the processing in step S307 in FIG. 18 (step S807).

(300) Next, the control module 151 sets and changes the speed of the conveyance roller 119, based on the detected thickness of the medium (step S808). The speed of the conveyance roller 119 is set within a range higher than or equal to the initial speed W1 and lower than or equal to the final speed W3a. In particular, the control module 151 sets the rotation speed of the conveyance roller 119 in such a way that the speed of the conveyance roller 119 is higher than or equal to the speed of the feed roller 112. Consequently, the control module 151 can suppress occurrence of jamming of a medium due to the medium being pushed by the feed roller 112 and being bent between the feed roller 112 and the conveyance roller 119.

(301) The speed of the conveyance roller 119 is set to increase as the thickness of a medium decreases and decrease as the thickness of the medium increases. The speed of the conveyance roller 119 may be set to increase as the thickness of the medium increases and decrease as the thickness of the medium decreases. The medium conveying apparatus 100 previously stores a table defining a relation between the thickness of a medium and the speed of the conveyance roller 119 into the storage device 140. The control module 151 specifies a speed related to the detected thickness with reference to the table stored in the storage device 140.

(302) Thus, the control module 151 sets the speed of the conveyance roller 119, based on a detected thickness of a medium. Consequently, the control module 151 can convey a medium at a suitable speed based on the thickness of the medium.

(303) As illustrated in the graph G103 in FIG. 36, the control module 151 changes the speed of the conveyance roller 119 to the speed set in step S808 while the feed roller 112 is stopped or decelerated.

(304) Next, the control module 151 resumes rotation of the feed roller 112 and changes the speed of the brake roller 113. Alternatively, the control module 151 changes the speeds of the feed roller 112 and the brake roller 113 (step S809).

(305) As illustrated in the graph G101 in FIG. 36, after setting the speed of the feed roller 112 to the final speed V3a, the control module 151 resumes rotation of the feed roller 112 or changes the speed of the feed roller 112 in such a way that the speed increases to the final speed V3a. Further,

as illustrated in the graph G102, the control module 151 changes the speed of the brake roller 113 to increase to the final speed U3a.

(306) Thus, when the front edge of the medium passes the fifth medium sensor 118, the control module 151 temporarily stops or decelerates the feed roller 112, detects the thickness of the medium, and sets the rotation speed of the conveyance roller 119, based on the thickness of the medium. Subsequently, the control module 151 resumes rotation of the feed roller 112 or accelerates the feed roller 112. When the thickness of the medium is detected during medium feeding, a detection error may occur due to, for example, shaking of the medium. By detecting the thickness of the medium in a state of temporarily stopping or decelerating the feed roller 112, the control module 151 can more precisely detect the thickness of the medium.

(307) After waiting until the front edge of the medium passes the position of the fifth medium sensor 118 in step S821, the control module 151 controls the first motor 131 to temporarily stop the feed roller 112 or reduce the speed of the feed roller 112 (step S822). From this point, the processing in steps S807 to S809 is executed, the control module 151 detects the thickness of the next medium, sets the rotation speed of the conveyance roller 119 based on the thickness of the medium, and subsequently resumes rotation of the feed roller 112 or accelerates the feed roller 112.

(308) As illustrated in the graph G101 in FIG. 36, the control module 151 stops or decelerates the feed roller 112 at the time T5. Subsequently, the control module 151 changes the speed of the conveyance roller 119 to the speed newly set in step S808 as illustrated in the graph G103 and resumes rotation of the feed roller 112 or accelerates the feed roller 112 as illustrated in the graph G101.

(309) As described in detail above, the medium conveying apparatus can control medium feeding better when changing the speed of the conveyance roller 119 based on the thickness of the medium.

(310) The medium conveying apparatus may drive the brake roller 113 by either the first motor or the second motor when changing the speed of the conveyance roller 119, based on the thickness of the medium. In that case, the control module 151 adjusts the timing of changing the speed of the brake roller 113 to the timing of changing the speed of the feed roller 112 or the conveyance roller 119. Further, the control module 151 may change the speed of the feed roller 112 or the brake roller 113 by controlling the first electromagnetic clutch 133 or the second electromagnetic clutch 134. Further, the medium conveying apparatus may include a driving force interruption mechanism, such as an electromagnetic clutch, between the conveyance roller 119 and the second motor, and the control module 151 may change the speed of the conveyance roller 119 by controlling the driving force interruption mechanism. The driving force interruption mechanism may be a planetary gear mechanism. The planetary gear is provided to be able to move between a position at which the planetary gear transmits the driving force from the second motor to the conveyance roller 119 and a position at which the planetary gear does not transmit the driving force from the second motor to the conveyance roller 119.

(311) FIG. 37 is a diagram illustrating a schematic configuration of an example processing circuit in a medium conveying apparatus according to another embodiment. The processing circuit 250 is used in place of the processing circuit 150 in the medium conveying apparatus 100 and executes the medium reading process, etc., in place of the processing circuit 150. The processing circuit 250 includes a control circuit 251, a determination circuit 252, etc. Each component may be independently configured with an integrated circuit, a microprocessor, firmware, etc.

(312) The control circuit 251 is an example of a control module and has a function similar to that of the control module 151. The control circuit 251 receives the operation signal from the operation device 105 or the interface device 135. Further, the control circuit 251 receives the first medium signal, the second medium signal, the ultrasonic signal, the fifth medium signal, and the sixth medium signal from the first medium sensor 111, the second medium sensor 114, the ultrasonic sensor 115, the fifth medium sensor 118, and the sixth medium sensor 121, respectively. Further, the control circuit 251 receives a determination result of a skew of a medium from the

determination circuit **252**. The control circuit **251** controls the first motor **131**, the second motor **132**, the third motor **336**, the first electromagnetic clutch **133**, and the second electromagnetic clutch **134**, based on the received information, acquires the input image from the imaging device **122**, and outputs the acquired image to the interface device **135**.

(313) The determination circuit **252** is an example of a determination module and has a function similar to that of the determination module **152**. The determination circuit **252** receives the third medium signal, the fourth medium signal, and the fifth medium signal from the third medium sensor **116**, the fourth medium sensor **117**, and the fifth medium sensor **118**, respectively. The determination circuit **252** determines whether a skew of a medium has occurred based on the received signals, and outputs the determination result to the control circuit **251**.

(314) As described in detail above, the medium conveying apparatus can control medium feeding better when the processing circuit **250** is used as well.

(315) While preferred embodiments have been described above, the embodiment is not limited thereto. For example, the medium conveying apparatus **100** may have a thin paper conveyance mode for conveying thin paper as a medium and a normal mode for conveying other media. In that case, the control module **151** sets the rotation speed of the feed roller **112** in such a way that the speed of the feed roller **112** in the normal mode is higher than the speed of the feed roller **112** in the thin paper conveyance mode. Consequently, thin paper is conveyed at a lower speed compared with other media, and therefore the control module **151** can suppress occurrence of damage to thin paper due to conveyance.

(316) Similarly to the aforementioned embodiments, the control module **151** sets the rotation speed of the feed roller **112** and the rotation speed of the conveyance roller **119** in such a way that the speed of the conveyance roller **119** is higher than the speed of the feed roller **112** in the normal mode and the thin paper conveyance mode. Consequently, the control module **151** can suppress occurrence of jamming of a medium due to the medium being pushed by the feed roller **112** and being bent between the feed roller **112** and the conveyance roller **119** in the normal mode and the thin paper conveyance mode. Furthermore, the control module **151** sets the rotation speed of the feed roller **112** and the rotation speed of the conveyance roller **119** in such a way that the ratio of the speed of the feed roller **112** to the speed of the conveyance roller **119** in the normal mode is less than the ratio of the speed of the feed roller **112** to the speed of the conveyance roller **119** in the thin paper conveyance mode. Consequently, the speed of the feed roller **112** approaches the speed of the conveyance roller **119** in the thin paper conveyance mode, and thin paper is more stably conveyed.

(317) The medium conveying apparatus may have a separation mode that separates and feeds a medium and a non-separation mode that feeds a medium without separation. In that case, when operating in the separation mode, the control module **151** executes each type of medium reading process described above. When operating in the non-separation mode, the control module **151** controls each motor in such a way that the brake roller **113** rotates in the medium feeding direction or rotates following the feed roller **112**. In that case, the control module **151** changes the magnitude of torque exerted on the brake roller **113** by controlling the second electromagnetic clutch **134**. Consequently, the control module **151** can suitably feed a medium in both the separation mode and the non-separation mode.

(318) The medium conveying apparatus may not have the high-speed mode, the medium-speed mode, and the low-speed mode and may operate in the high-speed mode in a fixed manner.

(319) In determination using the fifth medium sensor **118**, the control module **151** may use the second medium sensor **114** in place of the fifth medium sensor **118**. In other words, in each type of processing described above, the control module **151** may execute each type of processing executed when the front edge or the rear edge of a medium passes the position of the fifth medium sensor **118** at a time when the front edge or the rear edge of the medium passes the position of the second medium sensor **114**.

(320) The size or the thickness of a medium may be set by a user by using the operation device **105** or the information processing apparatus instead of being detected by using a sensor. Alternatively, the size or the thickness of a medium may be specified from a medium type (such as paper, a postcard, or a business card) set by a user by using the operation device **105** or the information processing apparatus. In that case, the medium conveying apparatus previously stores a table defining a relation between a medium type and the size or the thickness of the medium into the storage device **140**. The control module **151** specifies the size or the thickness of a conveyed medium with reference to the table stored in the storage device **140**.

(321) According to the embodiments, the medium conveying apparatus, the control method, and the control program can control medium feeding better.

(322) All examples and conditional language recited herein are intended for pedagogical purposes to aid the reader in understanding the invention and the concepts contributed by the inventor to furthering the art, and are to be construed as being without limitation to such specifically recited examples and conditions, nor does the organization of such examples in the specification relate to a presentation of the superiority and inferiority of the invention. Although the embodiment(s) of the present inventions have been described in detail, it should be understood that the various changes, substitutions, and alterations could be made hereto without departing from the spirit and scope of the invention.

Supplementary Note 1

(323) A medium conveying apparatus comprising: a loading tray to place media; a feed roller to separate and sequentially feed the media placed on the loading tray; a motor to drive the feed roller; a conveyance roller to convey the media fed by the feed roller; a sensor located between the feed roller and the conveyance roller and configured to detect a medium; and a control module to control the motor, wherein, in a separation period from a start of medium feeding by the feed roller to detection of a front edge of the first medium by the sensor, when feeding a first medium of media placed on the loading tray, the control module controls the motor to rotate the feed roller at a constant speed, and when feeding a second or subsequent medium, the control module controls the motor to rotate the feed roller at a first speed and then at a second speed higher than the first speed.

Supplementary Note 2

(324) The medium conveying apparatus according to supplementary note 1, wherein, when a front edge of a medium passes the conveyance roller, the control module controls the motor to continue rotation of the feed roller while reducing a rotation speed of the feed roller.

Supplementary Note 3

(325) The medium conveying apparatus according to supplementary notes 1 or 2, wherein the control module reduces a rotation speed of the conveyance roller when a rear edge of a medium passes the sensor.

Supplementary Note 4

(326) The medium conveying apparatus according to any one of supplementary notes 1 to 3, further comprising an imaging device located on a downstream side of the conveyance roller in a medium conveying direction and configured to image the media conveyed by the conveyance roller, wherein the control module increases a rotation speed of the conveyance roller when a rear edge of a medium passes an imaging position of the imaging device.

Supplementary Note 5

(327) The medium conveying apparatus according to any one of supplementary notes 1 to 4, wherein the control module increases a rotation speed of the feed roller when a rear edge of a medium passes the conveyance roller.

Supplementary Note 6

(328) The medium conveying apparatus according to any one of supplementary notes 1 to 5, wherein the control module detects a size of a medium, wherein when the size of a preceding medium is less than or equal to a size threshold value, the control module sets the second speed to a

speed lower than the second speed when the size of the preceding medium is greater than the size threshold value.

Supplementary Note 7

(329) The medium conveying apparatus according to any one of supplementary notes 1 to 6, wherein the control module detects a thickness of a medium, wherein when the thickness of a preceding medium is less than or equal to a thickness threshold value, the control module sets the second speed to a speed lower than the second speed when the thickness of the preceding medium is greater than the thickness threshold value.

Supplementary Note 8

(330) The medium conveying apparatus according to any one of supplementary notes 1 to 7, further comprising an ejection roller to eject the media conveyed by the conveyance roller, wherein the control module detects a size of a medium, wherein when the size of the medium is greater than a first size threshold value, the control module sets a rotation speed of the ejection roller to a speed lower than the rotation speed of the ejection roller when the size of the medium is less than or equal to the first size threshold value.

Supplementary Note 9

(331) The medium conveying apparatus according to any one of supplementary notes 1 to 7, further comprising an ejection roller to eject the media conveyed by the conveyance roller, wherein the control module detects a size of a medium, wherein when the size of the medium is less than a second size threshold value, the control module sets a rotation speed of the ejection roller to a speed lower than the rotation speed of the ejection roller when the size of the medium is greater than or equal to the second size threshold value.

Supplementary Note 10

(332) The medium conveying apparatus according to any one of supplementary notes 1 to 9, wherein the control module detects a size of a medium, and increases a rotation speed of the feed roller after the sensor detects a front edge of the medium, wherein when the size of a preceding medium is greater than a first size threshold value, the control module delays a timing of increasing a rotation speed of the feed roller relative to a timing of increasing the rotation speed of the feed roller when the size of the preceding medium is less than or equal to the first size threshold value.

Supplementary Note 11

(333) The medium conveying apparatus according to any one of supplementary notes 1 to 10, wherein the control module detects a size of a medium, and stops the feed roller when a front edge of the medium passes the conveyance roller and resumes feeding by the feed roller after a rear edge of the medium passes a predetermined position, wherein when the size of a medium of which a rear edge has passed the predetermined position or a medium preceding the medium is greater than a first size threshold value, the control module delays a resumption timing of feeding by the feed roller relative to a resumption timing of feeding by the feed roller when the size of the medium of which the rear edge has passed the predetermined position or the medium preceding the medium is less than or equal to the first size threshold value.

Supplementary Note 12

(334) The medium conveying apparatus according to any one of supplementary notes 1 to 11, wherein the control module detects a thickness of a medium, and sets a rotation speed of the conveyance roller based on the thickness of the medium.

Supplementary Note 13

(335) The medium conveying apparatus according to supplementary note 12, wherein, when a front edge of the medium passes the sensor, the control module temporarily stops or decelerates the feed roller, detects the thickness of the medium, and sets a rotation speed of the conveyance roller and subsequently, the control module resumes rotation of the feed roller or accelerates the feed roller.

Supplementary Note 14

(336) The medium conveying apparatus according to any one of supplementary notes 1 to 13,

wherein the control module sets a rotation speed of the conveyance roller in such a way that a surface moving speed of the conveyance roller is higher than or equal to a surface moving speed of the feed roller.

Supplementary Note 15

(337) The medium conveying apparatus according to supplementary note 14, wherein the medium conveying apparatus has a normal mode and a thin paper conveyance mode, and wherein the control module sets a rotation speed of the feed roller and a rotation speed of the conveyance roller in such a way that a ratio of the surface moving speed of the feed roller to the surface moving speed of the conveyance roller in the normal mode is less than a ratio of the surface moving speed of the feed roller to the surface moving speed of the conveyance roller in the thin paper conveyance mode, and the surface moving speed of the feed roller in the normal mode is higher than the surface moving speed of the feed roller in the thin paper conveyance mode.

Supplementary Note 16

(338) The medium conveying apparatus according to any one of supplementary notes 1 to 15, wherein the control module temporarily stops or decelerates the feed roller when a time elapsed after a rear edge of a preceding medium passes a first position until a front edge of a succeeding medium passes a second position is less than or equal to a predetermined time.

Supplementary Note 17

(339) The medium conveying apparatus according to any one of supplementary notes 1 to 16, wherein the control module temporarily stops or decelerates the feed roller when a front edge of a medium succeeding a preceding medium passes the sensor and determines a timing of resuming rotation of the feed roller or a timing of accelerating the feed roller based on a time elapsed after a rear edge of the preceding medium passes a first position until a front edge of the succeeding medium passes a second position.

Claims

1. A medium conveying apparatus comprising: a loading tray to place media; a feed roller to separate and sequentially feed the media placed on the loading tray; a motor to drive the feed roller; a sensor located on a downstream side of the feed roller in a medium conveying direction to detect a medium; and a processor to control the motor, wherein in a separation period from a start of medium feeding by the feed roller to detection of a front edge of the first medium by the sensor, when feeding a first medium of media placed on the loading tray, the processor controls the motor to rotate the feed roller at a constant speed, and when feeding a second or subsequent medium, the processor controls the motor to rotate the feed roller at a first speed and then at a second speed higher than the first speed.
2. The medium conveying apparatus according to claim 1, wherein the processor controls the motor to rotate the feed roller at a speed lower than the first speed when feeding the first medium of the media placed on the loading tray.
3. The medium conveying apparatus according to claim 1, wherein the processor controls the motor to rotate the feed roller at a speed higher than the second speed after detecting a front edge of the second or subsequent medium by the sensor.
4. The medium conveying apparatus according to claim 1, further comprising a brake roller located to face the feed roller and configured to rotate in a direction opposite to a medium feeding direction, wherein when feeding the first medium of the media placed on the loading tray, the processor sets a rotation speed of the brake roller in the separation period to a speed lower than a rotation speed of the brake roller in the separation period when feeding the second or subsequent medium.
5. The medium conveying apparatus according to claim 4, wherein, when feeding the first medium of the media placed on the loading tray, the processor sets a rotation speed of the brake roller in

- such a way that a surface moving speed of the brake roller in the separation period is higher than $\frac{1}{2}$ of a surface moving speed of the feed roller in the separation period when feeding the first medium.
6. The medium conveying apparatus according to claim 1, further comprising a conveyance roller to convey the media fed by the feed roller, wherein the processor determines whether a skew of a medium has occurred, and wherein, when it is determined that the skew of the medium has occurred by the processor, the processor sets the second speed in such a way that a surface moving speed of the feed roller is lower than a surface moving speed of the conveyance roller, and when it is not determined that the skew of the medium has occurred by the processor, the processor sets the second speed in such a way that a surface moving speed of the feed roller is identical to a surface moving speed of the conveyance roller.
7. The medium conveying apparatus according to claim 1, further comprising: a brake roller located to face the feed roller and configured to rotate in a direction opposite to a medium feeding direction; a conveyance roller to convey the media fed by the feed roller; and a second motor to drive the brake roller and the conveyance roller.
8. The medium conveying apparatus according to claim 7, further comprising an electromagnetic clutch provided on a driving force transmission path from a driving source of the brake roller to the brake roller.
9. The medium conveying apparatus according to claim 1, further comprising: a brake roller located to face the feed roller and configured to rotate in a direction opposite to a medium feeding direction; and a conveyance roller to convey the media fed by the feed roller; wherein the motor drives the brake roller along with the feed roller, and the medium conveying apparatus further comprises a second motor to drive the conveyance roller.
10. The medium conveying apparatus according to claim 9, further comprising an interruption mechanism that can interrupt a driving force from the motor to the feed roller, wherein, when a front edge of a medium passes the conveyance roller, the processor controls the interruption mechanism to continue transmitting a driving force from the motor to the brake roller while interrupting a driving force from the motor to the feed roller.
11. The medium conveying apparatus according to claim 1, further comprising: a brake roller located to face the feed roller and configured to rotate in a direction opposite to a medium feeding direction; a conveyance roller to convey the media fed by the feed roller; a second motor to drive the conveyance roller; and a third motor to drive the brake roller.
12. The medium conveying apparatus according to claim 1, further comprising a second sensor located between the feed roller and the sensor and configured to detect a medium, wherein the processor controls the motor to rotate the feed roller at the second speed when the second sensor detects a front edge of a medium in the separation period.
13. A method for conveying a medium, the method comprising: separating and sequentially feeding media placed on a loading tray by a feed roller; driving the feed roller by a motor; detecting a medium by a sensor located on a downstream side of the feed roller in a medium conveying direction; and controlling the motor, wherein in a separation period from a start of medium feeding by the feed roller to detection of a front edge of the first medium by the sensor, when feeding a first medium of media placed on the loading tray, controlling the motor to rotate the feed roller at a constant speed, and when feeding a second or subsequent medium, controlling the motor to rotate the feed roller at a first speed and then at a second speed higher than the first speed.
14. The method according to claim 13, wherein the motor is controlled to rotate the feed roller at a speed lower than the first speed when feeding the first medium of the media placed on the loading tray.
15. The method according to claim 13, wherein the motor is controlled to rotate the feed roller at a speed higher than the second speed after detecting a front edge of the second or subsequent medium by the sensor.
16. The method according to claim 13, wherein, when feeding the first medium of the media placed

on the loading tray, a rotation speed in the separation period of a brake roller located to face the feed roller and configured to rotate in a direction opposite to a medium feeding direction is set to a speed lower than a rotation speed in the separation period of the brake roller when feeding the second or subsequent medium.

17. A computer-readable, non-transitory medium storing executable instructions for conveying a medium, the executable instructions comprising: controlling a motor to drive a feed roller to separate and sequentially feed media placed on a loading tray, wherein in a separation period from a start of medium feeding by the feed roller to detection of a front edge of the first medium by a sensor located on a downstream side of the feed roller in a medium conveying direction, when feeding a first medium of media placed on the loading tray controlling the motor to rotate the feed roller at a constant speed, and when feeding a second or subsequent medium, controlling the motor to rotate the feed roller at a first speed and then rotate the feed roller at a second speed higher than the first speed.

18. The computer-readable, non-transitory medium according to claim 17, wherein the motor is controlled to rotate the feed roller at a speed lower than the first speed when feeding the first medium of the media placed on the loading tray.

19. The computer-readable, non-transitory medium according to claim 17, wherein the motor is controlled to rotate the feed roller at a speed higher than the second speed after detecting a front edge of the second or subsequent medium by the sensor.

20. The computer-readable, non-transitory medium according to claim 17, wherein, when feeding the first medium of the media placed on the loading tray, a rotation speed in the separation period of a brake roller located to face the feed roller and configured to rotate in a direction opposite to a medium feeding direction is set to a speed lower than a rotation speed in the separation period of the brake roller when feeding the second or subsequent medium.
